PeerJ Computer Science

A stable diffusion–based framework for multi-class stroke lesion segmentation in non-contrast brain CT

Ahmet Remzi Özcan

Department of Mechatronics Engineering, Bursa Technical University, Bursa, Turkey

ABSTRACT

Accurate segmentation of stroke subtypes in non-contrast computed tomography (CT) is essential for timely diagnosis and treatment planning. However, the inherently low tissue contrast in ischemic regions and the heterogeneous appearance of lesions across subtypes present significant challenges. This study introduces Stable Diffusion Multi-Class Segmentation (SDMC-Seg), a stable diffusion-based framework designed for multi-class stroke subtype segmentation in brain CT images. Unlike previous latent diffusion methods limited to binary tasks, SDMC-Seg addresses multi-class segmentation by encoding each stroke subtype into a distinct semantic channel, allowing the model to disentangle lesion-specific features in latent space for improved class-specific lesion delineation. The architecture integrates a fine-tuned mask autoencoder and a trainable CT image encoder, projecting both modalities into a shared, semantically meaningful latent space. Segmentation masks are reconstructed *via* a conditional denoising network operating on noisy latent representations conditioned on the CT image. In contrast to traditional diffusion methods requiring multiple iterative steps, SDMC-Seg performs single-step latent inference, enabling faster and more stable predictions while preserving structural detail. The model is evaluated on a curated, anonymized subset of a publicly available stroke CT dataset comprising 6,650 axial slices. Following common practice in lesion segmentation, quantitative training and evaluation are conducted on 2,223 stroke-positive slices (ischemic and hemorrhagic) to focus learning on lesion-specific patterns and to avoid dominance of background-only samples. It achieves an average Dice coefficient of 0.745 and an intersection over union (IoU) of 0.656, demonstrating consistently higher segmentation performance than U-Net, ResUNet, Attention U-Net, TransUNet, Swin-UNETR, and nnU-Net v2, particularly in ischemic stroke segmentation. Ablation results indicate that fine-tuning the mask autoencoder and jointly training the image encoder are both necessary; adapting both components yields the strongest gains. These results suggest that SDMC-Seg provides an efficient and scalable solution for stroke lesion delineation in CT, offering a computationally lightweight alternative to multi-step diffusion-based segmentation frameworks. Remaining limitations in detecting early ischemic changes highlight the need for future extensions incorporating multimodal imaging or temporal progression modeling.

Subjects Algorithms and Analysis of Algorithms, Artificial Intelligence, Computer Vision, Data Mining and Machine Learning, Neural Networks

Keywords Stroke lesion segmentation, Multi-class segmentation, Latent diffusion model, Single-step inference, Non-contrast CT

Submitted 25 September 2025
Accepted 16 March 2026
Published 27 April 2026

Corresponding author
Ahmet Remzi Özcan,
ahmet.ozcan@btu.edu.tr

Academic editor
Ankit Vishnoi

Additional Information and Declarations can be found on page 31

DOI **10.7717/peerj-cs.3823**

Copyright
2026 Özcan

Distributed under
Creative Commons CC-BY 4.0

OPEN ACCESS

How to cite this article Özcan AR. 2026. A stable diffusion–based framework for multi-class stroke lesion segmentation in non-contrast brain CT. **PeerJ Comput. Sci. 12:e3823 DOI 10.7717/peerj-cs.3823**

INTRODUCTION

A stroke is a medical emergency caused by blocked or bleeding blood vessels in the brain, resulting in severe physical and cognitive impairments. It is estimated that 12.2 million individuals worldwide experience a stroke annually, with approximately 6.5 million of these cases resulting in death. Over the past three decades, the global risk of stroke has increased by 70%, mainly due to the aging population (*Martin et al., 2024*). This constitutes a substantial challenge for healthcare systems on a global scale.

Strokes are classified into two primary categories: ischemic and hemorrhagic. An ischemic stroke, which accounts for approximately two-thirds of all cases, is caused by the partial or complete obstruction of a blood vessel supplying blood to the brain, resulting in oxygen deprivation in the affected tissue. Conversely, a hemorrhagic stroke is precipitated by the rupture of an arterial blood vessel, leading to cerebral hemorrhage. Despite their rarity, hemorrhagic strokes are associated with elevated mortality rates and more severe long-term outcomes, thus representing the most serious form of stroke (*Feigin et al., 2021*). In both types of stroke, the minimization of brain damage is contingent upon timely and accurate intervention.

Computed tomography (CT) and magnetic resonance imaging (MRI) findings are critical for diagnosing cerebrovascular disorders, including stroke. In patients presenting with stroke symptoms, identifying the presence and type of stroke is essential for ensuring an accurate classification and appropriate clinical management. Similarly, determining the location and size of the stroke is crucial for effective treatment planning. Despite the increasing availability of MRI, non-contrast brain CT remains the first-line imaging modality for stroke diagnosis due to its rapid acquisition time, widespread accessibility, and compatibility with emergency workflows (*Hollist et al., 2021*).

The manual interpretation of medical images by expert radiologists is a time-consuming and cognitively demanding task. Variability arising from differences in radiologists' experience levels and the limited availability of specialists, particularly in lower-tier healthcare settings, can lead to delays in diagnosis and treatment. These challenges have motivated the development of automated analysis methodologies based on machine learning techniques to enhance diagnostic efficiency and consistency.

Image processing techniques have been employed for decades to enhance and analyze medical images. However, traditional approaches are often insufficient for complex tasks such as precise segmentation, classification, and detection of pathological features. In recent years, advances in artificial intelligence and deep learning have offered significant potential for automated medical image analysis (*Zhou, Greenspan & Shen, 2023*). Among these methods, convolutional neural networks (CNNs) have demonstrated outstanding capabilities in feature extraction and disease classification across various imaging modalities, including chest X-rays, CT scans, and ultrasound images (*Liu et al., 2020*; *Frid-Adar et al., 2018*; *Schlemper et al., 2019*).

Despite their success, conventional CNN architectures are deficient in tasks requiring pixel-wise spatial prediction, such as segmentation, due to the loss of detailed spatial information in fully connected layers (*Sarvamangala & Kulkarni, 2022*). In response,

autoencoder-based architectures, most notably the U-Net architecture, address this challenge by introducing an encoder-decoder structure with skip connections that bridge low-level and high-level features, substantially improving the preservation of spatial information (*Ronneberger, Fischer & Brox, 2015*). U-Net and its extensions have become widely adopted in medical image segmentation, have demonstrated strong performance in tasks such as brain tumor segmentation in MRI scans (*Xue et al., 2018*) and lung nodule detection in CT scans (*Zhou et al., 2018*). Advanced models, such as U-Net++ and Attention U-Net (*Oktay et al., 2018*), have enhanced segmentation sensitivity by incorporating dense connections and attention mechanisms (*Siddique et al., 2021*). Building on the encoder-decoder paradigm, models such as ResUNet (*Zhang, Liu & Wang, 2018*) and SegResNet (*Myronenko, 2018*) incorporate residual connections to enhance feature propagation, stabilize gradient flow, and improve segmentation performance in complex medical imaging scenarios. Within the context of stroke lesion segmentation, similar encoder-decoder architectures have been extensively explored. *Kumar et al. (2020)* introduced CSNet, a dedicated convolutional framework for ischemic stroke lesion segmentation, demonstrating improved sensitivity to lesion boundaries in CT images. In a subsequent study, *Kumar et al. (2022)* proposed a lightweight asymmetric U-Net architecture tailored for acute ischemic stroke segmentation in CT perfusion (CTP) images, emphasizing computational efficiency while maintaining competitive segmentation accuracy. Among U-Net–based approaches, nnU-Net has emerged as a strong and widely adopted baseline by introducing a self-configuring segmentation framework that automatically adapts network architecture, preprocessing, and training hyperparameters to the target dataset. By eliminating manual design choices, nnU-Net and its updated version nnU-Net v2 have demonstrated state-of-the-art or near–state-of-the-art performance across a wide range of medical segmentation benchmarks, including brain lesion segmentation tasks (*Isensee et al., 2021*). Consequently, nnU-Net v2 is increasingly used as a reference baseline for fair and reproducible evaluation of newly proposed segmentation methods.

To overcome the limitations of convolutional architectures in modeling long-range dependencies and global contextual interactions, transformer-based models have recently gained prominence in medical image segmentation. Unlike purely convolutional networks that rely on local receptive fields, transformers employ self-attention mechanisms to capture long-range spatial relationships and high-level semantic context, which are particularly important for anatomically complex structures and diffuse lesion patterns. Representative examples include TransUNet (*Chen et al., 2021*), which integrates Vision Transformers into the U-Net encoder to combine global self-attention with localized feature decoding, and Swin-UNETR (*Hatamizadeh et al., 2021*), which leverages hierarchical window-based transformers for efficient modeling of multi-scale context. These approaches have demonstrated strong performance in challenging segmentation tasks, including multi-organ CT and brain MRI segmentation, where contextual reasoning plays a critical role.

In parallel, diffusion models have emerged as powerful generative frameworks with applications in image generation, super-resolution, inpainting, and medical image

segmentation. These models learn and reconstruct complex data distributions through an iterative noise perturbation and denoising process (*Sohl-Dickstein et al., 2015*). Denoising Diffusion Probabilistic Models (DDPMs) (*Ho, Jain & Abbeel, 2020*) represent a foundational class that incorporates Gaussian noise in the forward process within a probabilistic framework. In medical imaging, diffusion-based methods have shown promise in reconstruction, anomaly detection, segmentation, and denoising tasks (*Kazerouni et al., 2023*).

Segmentation methods based on image-level diffusion models, such as MedSegDiff, leverage image conditioning to generate context-aware segmentation predictions (*Wu et al., 2024a*; *Wolleb et al., 2022*; *Xing et al., 2023*). In this approach, Gaussian noise is progressively added to the segmentation mask during the forward process. In the reverse process, a denoising U-Net conditioned on the corresponding medical image is used to estimate and remove the noise, gradually reconstructing the segmentation output. This enables the generation of segmentation masks tailored to the structural characteristics of the input image. However, performing diffusion directly in image space incurs high computational costs, particularly for multi-dimensional and high-resolution images. Latent Diffusion Models (LDMs) address this challenge by shifting the diffusion process from pixel space to a lower-resolution latent space, thereby preserving generative performance while significantly reducing computational costs (*Rombach et al., 2022*). This is achieved by using pre-trained encoders to map high-resolution data into a semantically meaningful, lower-resolution latent representation.

The remarkable achievements of leading LDM-based models, such as Stable Diffusion (*Rombach et al., 2022*), DALL-E 2 (*Ramesh et al., 2022*), and Imagen (*Saharia et al., 2022*), in areas such as text-to-image synthesis, image reconstruction, and creative design have inspired the development of segmentation techniques that adopt similar approaches, including LSegDiff (*Vu Quoc et al., 2023*) and LDSeg (*Zaman et al., 2024*). However, training these segmentation models from scratch without pre-trained weights results in protracted training times and substantial computational overhead, as the diffusion process typically requires dozens of iterative denoising steps. In addition, their inference phase involves numerous sampling steps, leading to prolonged processing times that limit their applicability in time-sensitive medical imaging settings. These challenges have driven recent research efforts toward optimizing diffusion-based segmentation models for improved efficiency and scalability. In this context, SDSeg (*Lin et al., 2024*) demonstrates the successful adaptation of the Stable Diffusion architecture for biomedical image segmentation, achieving promising results in single-class segmentation tasks. The optimized sampling process reduces inference complexity compared to earlier diffusion-based approaches, and the utilization of pre-trained weights enhances adaptability across various medical imaging modalities. Nevertheless, existing evaluations of SDSeg are restricted to single-class segmentation scenarios, and its design has not been explicitly validated for multi-class lesion delineation. Extending latent diffusion models to multi-class segmentation introduces additional challenges, as the shared latent space must preserve clear inter-class boundaries while maintaining intra-class coherence. This issue is particularly pronounced in brain stroke CT imaging, where ischemic and hemorrhagic

lesions may exhibit overlapping intensity distributions and ambiguous spatial characteristics. As a result, the direct application of single-class latent diffusion frameworks to multi-class stroke segmentation is non-trivial and remains insufficiently explored. Accordingly, SDMC-Seg (Stable Diffusion Multi-Class Segmentation) is proposed to extend the capabilities of Stable Diffusion (*Rombach et al., 2022*) for effective multi-class brain stroke segmentation, explicitly addressing class disentanglement between ischemic and hemorrhagic stroke regions within a unified latent diffusion framework. Unlike many existing diffusion-based segmentation approaches, which predominantly focus on binary or single-class tasks or rely on image-space diffusion, the proposed framework formulates multi-class stroke segmentation as a conditional latent diffusion problem. The main contributions of this work are summarized as follows:

- Segmentation masks and CT images are encoded into a low-resolution yet semantically rich latent space, enabling the diffusion process to operate with significantly lower computational complexity than conventional image-space methods. This reduces training time and resource usage, improving scalability for high-resolution and volumetric medical imaging tasks.
- The AutoencoderKL module, responsible for encoding segmentation masks, is fine-tuned for multi-class segmentation through a dedicated pre-training phase. This pre-training step ensures more precise and semantically aligned latent representations, leading to improved segmentation performance.
- A single-step latent estimation strategy is introduced to eliminate the need for iterative sampling in multi-class segmentation. While conceptually inspired by recent latent diffusion–based segmentation studies, the strategy is reformulated to support class-discriminative latent supervision and stable single-step inference in the multi-class setting.
- Improved segmentation performance is demonstrated compared to widely used models such as U-Net, ResUNet, Attention U-Net, TransUNet, SegResNet, Swin-UNETR, and nnU-Net v2, evaluated on a stroke CT dataset with expert-annotated ischemic and hemorrhagic lesion regions.

BACKGROUND

Denoising diffusion probabilistic model

Denoising Diffusion Probabilistic Models (DDPMs) represent a class of two-stage generative models that have garnered attention for their capacity to model data distributions. These models are trained using a framework of forward and reverse diffusion processes and are capable of producing highly accurate data samples in the inference stage. In the forward diffusion process, Gaussian noise is iteratively added to the input data, distorting it toward a pure noise distribution. Then, the reverse diffusion process removes this noise step by step, producing samples that are close to the original data distribution. This bidirectional structure enables DDPMs to learn complex data distributions effectively, leveraging the principles of Markovian processes and variational inference.

In the mathematical formulation of the forward diffusion process, a data sample x_0 is transformed into a sequence of noisy versions $x_1, x_2, \ldots, x_t$ over T timesteps by iteratively adding Gaussian noise. This process can be described as follows:

$$q(x_t|x_{t-1}) := \mathcal{N}(x_t; \sqrt{1-\beta_t}x_{t-1}, \beta_t I) \tag{1}$$

where β_t is a predefined variance schedule controlling the noise magnitude at each step. As noise propagates over multiple steps, its cumulative effect can be described as a closed-form solution:

$$q(x_t|x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t}x_0, (1-\bar{\alpha}_t)I) \tag{2}$$

where $\alpha_t := 1 - \beta_t$ and $\bar{\alpha}_t := \prod_{s=1}^{t} \alpha_s$ represents the cumulative product of noise factors up to time. A structured transformation allows x_t to be explicitly formulated as a function of x_0, enabling the direct computation of noisy samples from the original data.

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t} \cdot \varepsilon, \quad \text{where } \varepsilon \sim \mathcal{N}(0, I). \tag{3}$$

The reverse diffusion process aims to recover the original data distribution by gradually denoising the noisy sample x_T, starting from a Gaussian prior $x_T \sim \mathcal{N}(0, I)$. Formally, the reverse sampling process is defined as follows:

$$p_\theta(x_{t-1}|x_t) := \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t)) \tag{4}$$

where $\mu_\theta(x_t, t)$ represents the predicted mean of the denoised sample at timestep t, and $\Sigma_\theta(x_t, t)$ denotes the covariance matrix, modeling the uncertainty of the prediction. The predicted mean $\mu_\theta(x_t, t)$ can be expressed as:

$$\mu_\theta(x_t, t) = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}t}}\varepsilon_\theta(x_t, t)\right). \tag{5}$$

Accordingly, x_{t-1} is inferred from x_t using the following formulation, as presented in *Ho, Jain & Abbeel (2020)*:

$$x_{t-1} = \mu_\theta(x_t, t) + \sigma_t z, \quad z \sim \mathcal{N}(0, I) \tag{6}$$

where σ_t defines the variance scheme, which can be either pre-defined or learned by model, as proposed by *Dhariwal & Nichol (2021)*. This formulation highlights the stochastic nature of the sampling process due to the presence of the random component z. To estimate the noise component at each timestep t, the noise predictor network $\varepsilon_\theta(x_t, t)$ is employed as an adapted form of the U-Net architecture. The predicted noise $\varepsilon_\theta(x_t, t)$ is learned by the model to iteratively refine the denoised samples and reconstruct x_0. Thus, the training objective is formulated as a simple mean-squared error (MSE) loss between the predicted noise and the true noise as follows:

$$L = E_{t,x_0,\varepsilon}\left[|\varepsilon - \varepsilon_\theta(x_t, t)|^2\right]. \tag{7}$$

Diffusion models for medical image segmentation

Medical image segmentation aims to generate a structured representation by delineating anatomical structures and assigning categorical labels to different regions in an image.

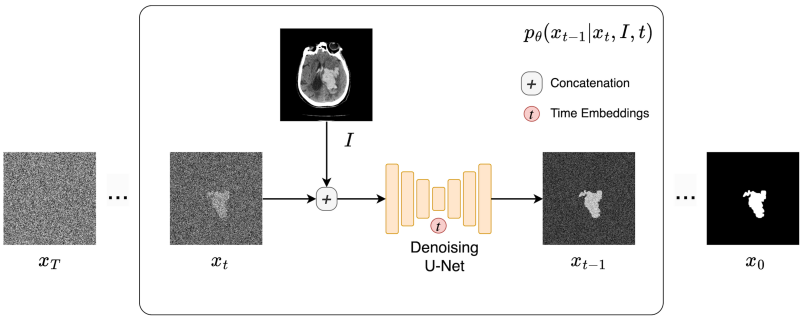


Figure 1 Illustration of the conditioning process during the reverse step in diffusion-based brain stroke segmentation performed in pixel space. Full-size DOI: 10.7717/peerj-cs.3823/fig-1

Conventional deep learning-based methods, including U-Net and its derivatives, employ discriminative training to establish direct mappings from input images to segmentation masks. Conversely, diffusion models address segmentation as a generative denoising process, wherein a segmentation mask is progressively reconstructed from a noisy representation. In diffusion-based segmentation, the forward process iteratively adds noise to the input mask, while the reverse process estimates the noise component to reconstruct the original segmentation mask. Unlike conventional diffusion models used for image generation, segmentation models are conditioned on the image to be segmented during the reverse process. This allows the model to integrate the generative capability of diffusion with the structural consistency of the input image, ensuring anatomically coherent segmentation. Figure 1 illustrates the conditioning process during the reverse step, demonstrating its application in brain stroke segmentation performed in pixel space.

Initial efforts in diffusion-based medical image segmentation explored probabilistic approaches to iteratively refine segmentation masks. *Wolleb et al. (2022)* demonstrated that diffusion models could generate multiple plausible segmentation masks by leveraging an ensemble-based approach. By incorporating uncertainty estimation, this method provided more reliable segmentation results, particularly in cases where anatomical structures exhibited high variability. *Guo et al. (2023)* introduced a pre-segmentation diffusion sampling technique in which an auxiliary segmentation network generates the initial estimate prior to the diffusion process. This approach significantly reduced computational costs by minimizing the number of denoising steps required. Building on these advancements, *Wu et al. (2024a)* introduced MedSegDiff, incorporating the Feature Frequency Parser (FF-Parser) to eliminate the negative effects of high-frequency components and enhance segmentation accuracy. Additionally, the dynamic conditional encoding strategy is introduced for step-wise attention, enhancing structural consistency across multiple imaging modalities (*Wu et al., 2024a*). MedSegDiff-V2 (*Wu et al., 2024b*) integrates Vision Transformers to capture long-range dependencies, enhancing segmentation consistency and adaptability. Diff-UNet (*Xing et al., 2023*) extends diffusion-based segmentation to 3D medical imaging by integrating a diffusion model with a U-Net architecture, aiming to improve semantic feature extraction and volumetric segmentation accuracy. By incorporating stepwise feature fusion, it enhances multi-scale

feature processing and enables more effective modeling of long-range dependencies in volumetric segmentation tasks. BerDiff (*Chen, Wang & Shan, 2023*) introduces a Bernoulli-based diffusion model for medical image segmentation, diverging from conventional Gaussian noise diffusion. By employing a conditional Bernoulli diffusion process, it seeks to improve segmentation accuracy, especially in low-contrast and binary segmentation tasks, while maintaining anatomical coherence.

Diffusion-based approaches operating in pixel space have been shown to be effective in segmentation tasks by applying the diffusion process directly to segmentation masks. However, these approaches are confronted with challenges such as high resource demands, long inference times, and limitations in capturing global contextual relationships. To address these challenges, latent-space diffusion models have emerged as a promising alternative. By performing the diffusion process within a compact, semantically rich latent representation, these models enhance computational efficiency and structural consistency while preserving critical anatomical details. LSegDiff (*Vu Quoc et al., 2023*) exemplifies this approach by leveraging a Variational Autoencoder (VAE) (*Kingma & Welling, 2019*) to encode segmentation masks into a compact latent space before applying the diffusion process. After denoising in the latent space, a VAE decoder reconstructs the segmentation mask, ensuring high anatomical fidelity. Experimental evaluations on polyp and skin lesion segmentation datasets demonstrate that LSegDiff improves segmentation accuracy while enhancing computational efficiency compared to pixel-space diffusion models. LDSeg (*Zaman et al., 2024*), a latent diffusion-based segmentation model, employs a Mask Autoencoder for latent transformation, aiming to reduce information loss and enhance reconstruction fidelity. It incorporates a structured noise injection strategy in the latent space to mitigate artifacts and improve segmentation robustness. Evaluations on diverse medical imaging datasets indicate its effectiveness in handling heterogeneous and noisy data. Beyond these latent-space diffusion approaches, SDSeg (*Lin et al., 2024*) adapts the Stable Diffusion framework for medical image segmentation, leveraging its learned representations to shorten training time and enhance generalization across diverse medical imaging modalities. It employs a single-step reverse diffusion process, reducing the number of denoising iterations required during inference and improving computational efficiency compared to conventional multi-step diffusion-based segmentation frameworks. While SDSeg has demonstrated its effectiveness in single-class segmentation tasks, its potential for multi-class segmentation remains an open area for further investigation.

Denoising diffusion implicit model

Denoising Diffusion Probabilistic Model (DDPM) provides a powerful framework for modeling complex data distributions through a stepwise denoising process. However, one of its key limitations is the computational cost associated with the large number of timesteps required for sampling. Since DDPM relies on a Markovian reverse diffusion process, reconstructing an image from pure Gaussian noise typically involves thousands of sequential steps, making it inefficient for applications requiring fast inference and high-resolution generation.

To address this inefficiency, the Denoising Diffusion Implicit Model (DDIM) (*Song, Meng & Ermon, 2020*) is introduced as a more efficient sampling method. Unlike DDPM, DDIM removes the strict Markovian assumption, allowing non-Markovian updates that enable the model to skip multiple timesteps in the reverse diffusion process. This approach significantly reduces the number of required steps while maintaining the quality of the generated samples, making it computationally more efficient.

In DDIM, the reverse diffusion process is reformulated to first estimate a denoised latent representation, $\hat{x}_0$, rather than directly computing x_{t-1}. This predicted latent sample then serves as an intermediate step for computing x_{t-1}, resulting in a more structured and computationally efficient denoising trajectory. The estimation of $\hat{x}_0$ is given by:

$$\hat{x}_0 = \frac{1}{\sqrt{\bar{\alpha}_t}} \left(x_t - \sqrt{1 - \bar{\alpha}_t} \cdot \varepsilon_\theta(x_t, t) \right). \tag{8}$$

Using $\hat{x}_0$, the updated sample at timestep $t - 1$ is then computed as:

$$x_{t-1} = \sqrt{\bar{\alpha}_{t-1}} \cdot \hat{x}_0 + \sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2} \cdot \varepsilon_\theta(x_t, t) + \sigma_t \cdot z \tag{9}$$

where σ_t controls the level of stochasticity in the sampling process, and $z \sim \mathcal{N}(0, I)$ when stochastic sampling is applied. By setting $\sigma_t = 0$, the process becomes fully deterministic, ensuring consistent sample generation across multiple runs. This formulation allows DDIM to retain compatibility with the DDPM training framework while significantly accelerating inference by requiring only a fraction of the original sampling steps.

PROPOSED METHODOLOGY

In this work, a multi-class brain stroke segmentation approach leveraging Stable Diffusion in latent space is proposed to perform lesion-level segmentation of ischemic and hemorrhagic stroke regions, where non-stroke tissue is implicitly treated as background. Here, the term multi-class refers to the simultaneous segmentation of multiple stroke lesion subtypes (ischemic and hemorrhagic) within a single framework, rather than image-level stroke detection or classification. Building upon the framework introduced by *Lin et al. (2024)*, the proposed method is adapted for multi-class segmentation in brain CT images. To this end, the latent encoding process is refined to effectively represent the multi-class structure of stroke segmentation. Both CT images and segmentation masks are encoded into a low-dimensional latent space, allowing an efficient denoising diffusion process. The conditional latent denoiser learns to estimate the noise added at each time step by conditioning on image embeddings from the CT image, thereby enabling the reconstruction of the segmentation mask in the reverse process. In the inference stage, a single-step sampling strategy is employed to reduce the number of reverse diffusion iterations required for segmentation, while maintaining high segmentation accuracy. The proposed methodology consists of three main components: latent space encoding, conditional latent denoising, and reverse process in inference. The overall architecture of the proposed model is illustrated in Fig. 2.

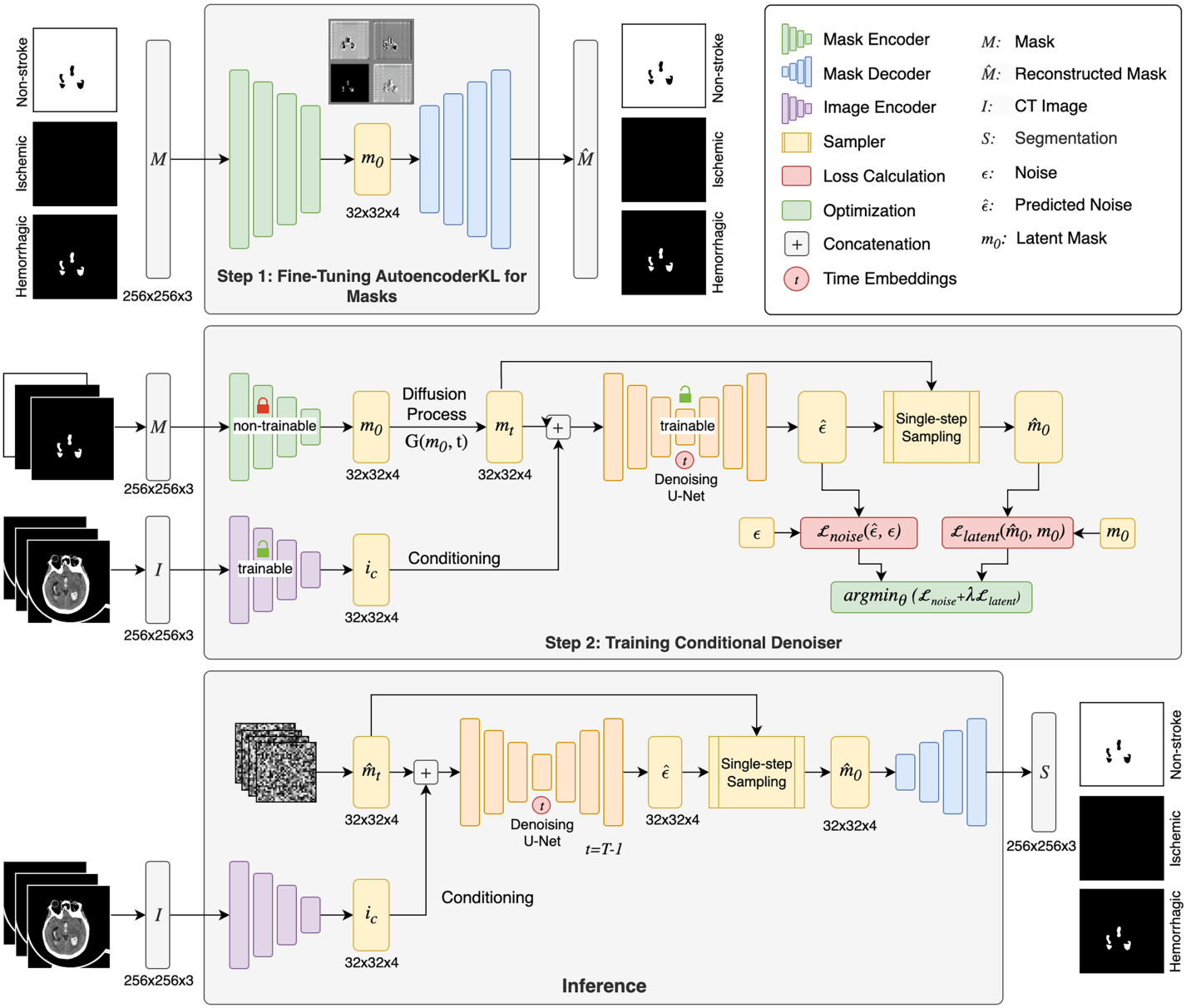


Figure 2 **Proposed multi-class Stable Diffusion segmentation model.** The model is trained in two stages: (1) mask autoencoder fine-tuning, where an AutoencoderKL is optimized to encode multi-class segmentation masks into compact latent representations. (2) Conditional latent denoising, where a denoising network is trained to predict the added noise at each timestep, conditioned on the image embedding. During inference, a single-step reverse process is employed to reconstruct the segmentation mask from the latent representation.

Full-size DOI: 10.7717/peerj-cs.3823/fig-2

Latent space encoding

Latent space encoding transforms high-resolution medical images and their segmentation masks into a compact, semantically rich representation for efficient processing and reconstruction. This is achieved using an encoder network that maps the input data into a low-resolution latent space while preserving essential structural features. To ensure accurate and reliable segmentation, the encoder network should be optimised to retain

critical details from both medical images and segmentation masks in the latent representation.

In Stable Diffusion, latent space encoding is performed using a VAE. The encoder network $\mathscr{E}$ maps input images $x \in \mathbb{R}^{H\times W\times 3}$ in RGB space to a latent representation $z = \mathscr{E}(x)$, reducing the spatial resolution while preserving key structural features. The decoder network $\mathscr{D}$ reconstructs the image from its latent representation, producing $\hat{x} = \mathscr{D}(z) = \mathscr{D}(\mathscr{E}(x))$, where $z \in \mathbb{R}^{h\times w\times c}$. A downsampling factor $f = H/h = W/w$ is applied to control the compression level. To regularize the latent space, two techniques are used: KL-Regularization (KL-reg.), which applies a Kullback-Leibler (KL) divergence penalty to ensure that the latent distribution remains close to a standard normal prior; and Vector Quantization Regularization (VQ-reg.), which discretizes the latent space using a codebook, as in VQ-VAE (*van den Oord, Vinyals & Kavukcuoglu, 2017*). These techniques contribute to an efficient latent space by reducing redundancy and optimizing computational efficiency for improved segmentation accuracy.

A typical 2D medical segmentation dataset consists of pairs of medical images $I \in \mathbb{R}^{H\times W\times 1}$ and their corresponding segmentation masks $M \in \mathbb{R}^{H\times W\times 1}$. For single-class segmentation tasks, masks are represented as binary images, where each pixel indicates the presence or absence of the target structure. In contrast, for multi-class segmentation, masks are encoded using distinct numerical values, with each value corresponding to a different anatomical or pathological class. In the training process of the proposed model, segmentation masks are utilized as inputs, while medical images are employed for conditioning, as illustrated in Fig. 2. The autoencoder utilized by Stable Diffusion, originally trained on RGB images, is inadequate for generating a representative latent space for multi-class segmentation masks. To address this limitation, the AutoencoderKL is fine-tuned on multi-class stroke segmentation masks in a dedicated pre-training stage. This adaptation encourages the latent space to encode categorical boundaries and class-specific morphology, rather than image-style statistics inherited from natural RGB training data. As a result, the downstream denoising network receives semantically aligned latent codes that better preserve ischemic-hemorrhagic separation and reduce inter-class mixing during reconstruction.

The medical images used for conditioning are also encoded into the latent space using the same encoder structure, but without an additional pre-training step. This design isolates domain adaptation to the mask latent space, where discrete label semantics must be preserved most strictly. In contrast, the image encoder can be optimized jointly with the denoiser to learn task-specific CT features during conditional training. Instead, the encoder remains trainable during the main training phase, allowing fine-tuning to capture the specific characteristics of medical images. In the pre-training stage, segmentation masks are represented using one-hot encoding, where each segmentation class constitutes a distinct channel in the input data. This approach is motivated by the fact that directly encoding segmentation masks using distinct numerical values could introduce artificial relationships and unintended semantic distances between classes in latent space. Utilizing one-hot encoding, each class is independently represented in its own dimension, thereby

enabling clearer differentiation among classes and enhancing the quality of latent representations. The mask encoder and decoder are denoted as $\mathscr{E}_{\text{mask}}$ and $\mathscr{D}_{\text{mask}}$, respectively. Given a one-hot encoded segmentation mask $M \in \mathbb{R}^{H\times W\times C}$, where C is the number of classes, the mask encoder transforms it into a compact latent representation $m_0 = \mathscr{E}_{\text{mask}}(M)$, where $m_0 \in \mathbb{R}^{h\times w\times c}$.

In the proposed method, latent encoding is performed using an AutoencoderKL architecture, which extends the variational autoencoder framework by incorporating KL-regularization to enforce a structured latent representation, making it particularly suitable for segmentation tasks. The training process incorporates a perceptual loss function based on image patch similarity and an adversarial discriminator to enhance reconstruction quality. The total loss function for the autoencoder is formulated as:

$$\mathscr{L}_{\text{AE}} = \lambda_{\text{rec}}\mathscr{L}_{\text{rec}} + \lambda_{\text{KL}}\mathscr{L}_{\text{KL}} + \lambda_{\text{adv}}\mathscr{L}_{\text{adv}} \tag{10}$$

where $\mathscr{L}_{\text{rec}}$ represents the combined reconstruction loss, consisting of pixel-wise (L1) and perceptual (patch similarity) components; $\mathscr{L}_{\text{KL}}$ denotes the KL divergence loss, which regularizes the latent space by encouraging the approximate posterior to align with a standard normal prior; and $\mathscr{L}_{\text{adv}}$ corresponds to the adversarial generator loss. The adversarial loss is incorporated after a certain number of iterations to ensure stable initial training, and its weight λ_{adv} is adaptively scaled during optimization. This combination facilitates reconstructions that align closely with perceptual quality, thereby improving the latent representations learned for segmentation masks. As visualized in Fig. 3, multi-class segmentation masks are more accurately reconstructed by the fine-tuned AutoencoderKL. The figure also includes the latent representations generated by the fine-tuned model, highlighting spatial structure and semantic class separation within the segmentation masks.

Conditional latent denoising

The proposed model performs segmentation through a diffusion process conducted in the latent space. In this process, latent representations of segmentation masks are corrupted with noise and subsequently restored using a denoising U-Net trained to predict the added noise conditioned on the corresponding input image. This strategy significantly reduces the computational cost compared to performing the diffusion process directly in pixel space, which is often redundant for segmentation tasks due to the limited semantic complexity of binary or categorical maps. To guide the denoising process with contextual information from the source image, a trainable image encoder is employed to generate a latent embedding of the input CT image. Although its architecture is identical to that of the segmentation mask encoder, the image encoder is initialized with pretrained weights from the original Stable Diffusion model and trained jointly with the denoising U-Net. The image encoder, denoted as $\mathscr{E}_{\text{image}}$, maps the input CT image I to a latent representation $i_c = \mathscr{E}_{\text{image}}(I)$, which is used to condition the denoising process. During training, the resulting image embedding is concatenated with the noisy latent mask representation at each diffusion timestep and provided as input to the denoising U-Net. The overall training

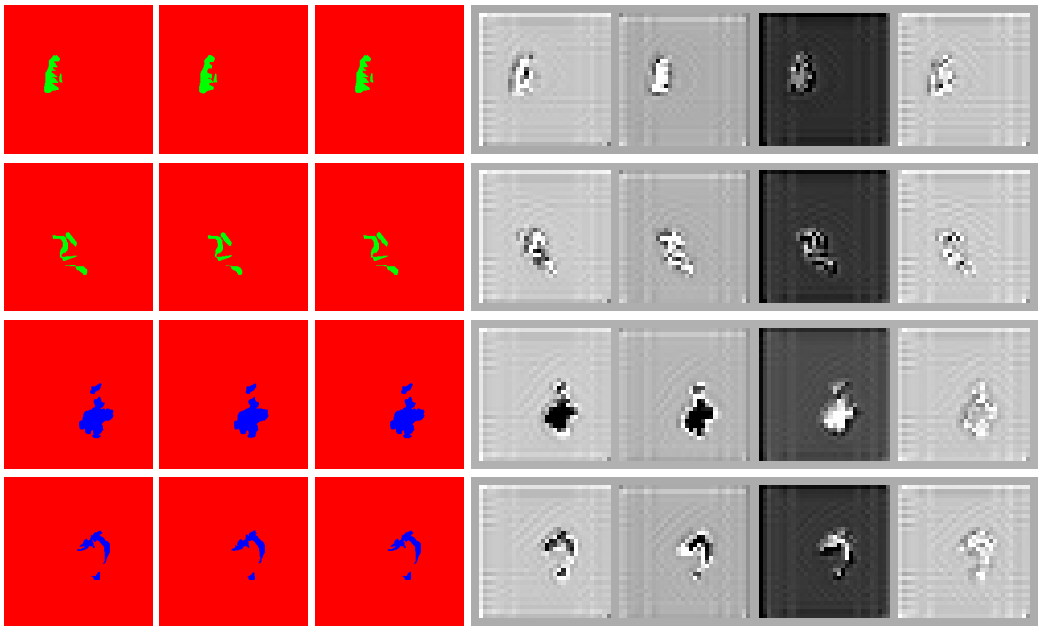


Figure 3 Qualitative comparison of reconstruction results produced by the original and fine-tuned AutoencoderKL models for multi-class segmentation masks. Reconstructions are obtained as $\hat{M} = \mathscr{D}_{\text{mask}}(m_0)$, where $m_0 = \mathscr{E}_{\text{mask}}(M)$ denotes the latent representation of the input mask. In all visualizations, red, green, and blue channels correspond to non-stroke, ischemic, and hemorrhagic regions, respectively. Full-size DOI: 10.7717/peerj-cs.3823/fig-3

procedure of the proposed conditional latent denoising framework is summarized in Algorithm 1.

In standard latent diffusion frameworks such as Stable Diffusion, cross-attention mechanisms are commonly employed to incorporate conditioning information, particularly in multimodal generation tasks. However, in image-to-image segmentation, where the objective is to extract modality-specific semantic and structural representations, the benefits of such mechanisms have been found to be limited. Moreover, the inclusion of cross-attention layers across multiple stages of the denoising U-Net introduces considerable computational overhead. Inspired by the lightweight conditioning strategy explored in SDSeg, a latent fusion approach based on channel-wise concatenation is employed in this study. Instead of attention-based fusion, the latent representations of the input image and the segmentation map are concatenated and provided to the denoising U-Net. This design choice enables computational efficiency while preserving structural coherence within the latent space. As illustrated in Fig. 3, strong spatial alignment is observed between latent features and their corresponding segmentation targets, supporting the effectiveness of this strategy in the context of brain stroke segmentation.

In the forward diffusion process, the initial latent representation of the segmentation map, denoted as m_0, is gradually noised through a Gaussian process over T timesteps. The

Algorithm 1 **Training procedure of SDMC-Seg (conditional latent denoising).**

Input: Mini-batch $\mathcal{B} = \{(I, M)\}$ where I is CT image and M is one-hot mask
Input: Mask encoder $\mathcal{E}_{\text{mask}}$, mask decoder $\mathcal{D}_{\text{mask}}$, image encoder $\mathcal{E}_{\text{image}}$, denoiser f_θ
Input: Diffusion steps T, noise schedule $\{\bar{\alpha}_t\}_{t=1}^{T}$, latent loss weight λ
Output: Updated denoiser parameters θ
foreach *mini-batch* $\mathcal{B}$ **do**
Encode segmentation mask into latent space;
$m_0 \leftarrow \mathcal{E}_{\text{mask}}(M)$;
Encode conditioning image;
$i_c \leftarrow \mathcal{E}_{\text{image}}(I)$;
Sample diffusion timestep and noise;
$t \sim \mathcal{U}\{1, \ldots, T\}$; $\quad \varepsilon \sim \mathcal{N}(0, I)$;
Forward noising in latent space;
$m_t \leftarrow \sqrt{\bar{\alpha}_t}\, m_0 + \sqrt{1 - \bar{\alpha}_t}\, \varepsilon$;
Denoiser prediction conditioned on image latent;
$\hat{\varepsilon} \leftarrow f_\theta(m_t, t; i_c)$;
Single-step estimate of clean latent;
$\hat{m}_0 \leftarrow \frac{1}{\sqrt{\bar{\alpha}_t}}(m_t - \sqrt{1 - \bar{\alpha}_t}\, \hat{\varepsilon})$;
Compute loss terms;
$\mathcal{L}_{\text{noise}} \leftarrow \|\hat{\varepsilon} - \varepsilon\|_1$;
$\mathcal{L}_{\text{latent}} \leftarrow \|\hat{m}_o - m_o\|_1$;
Total loss and parameter update;
$\mathcal{L} \leftarrow \mathcal{L}_{\text{noise}} + \lambda\, \mathcal{L}_{\text{latent}}$;
$\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}$;

noisy latent at timestep t, denoted as m_t, is obtained by applying a Gaussian noise function parameterized by a predefined variance schedule:

$$m_t = G(m_0, t) = \sqrt{\bar{\alpha}_t} \cdot m_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I) \tag{11}$$

where $\bar{\alpha}_t$ represents the cumulative product of the noise scaling coefficients, and ε is drawn from a standard Gaussian distribution. This process produces increasingly noisy latent representations, which are used as input to the denoising network during training. The denoising U-Net is trained to predict the added noise ε from the noisy latent m_t, conditioned on the image embedding i_c, such that $\hat{\varepsilon} = f_{\text{denoiser}}(m_t; i_c)$. Based on this prediction, the estimated latent representation at timestep zero, denoted as $\hat{m}_0$, is computed by inverting the forward process:

$$\hat{m}_0 = \frac{1}{\sqrt{\bar{\alpha}_t}} \left(m_t - \sqrt{1 - \bar{\alpha}_t} \cdot \hat{\varepsilon} \right). \tag{12}$$

To optimize the training process, a composite loss function is employed, combining both noise prediction and latent reconstruction objectives. The noise loss $\mathcal{L}_{\text{noise}}$ is defined as the mean absolute error (MAE) between the predicted noise $\hat{\varepsilon}$ and the true noise ε added during the forward process. In parallel, the latent loss $\mathcal{L}_{\text{latent}}$ measures the MAE between the estimated latent representation $\hat{m}_0$ and the original latent m_0. The total training loss is expressed as:

$$\mathcal{L} = \mathcal{L}_{\text{noise}} + \lambda \mathcal{L}_{\text{latent}} = \|\hat{\varepsilon} - \varepsilon\|_1 + \lambda \|\hat{m}_0 - m_0\|_1 \tag{13}$$

Algorithm 2 **Inference and evaluation of SDMC-Seg (single-step reverse).**

Input: Test/validation sample (I, M_{gt})
Input: Trained image encoder $\mathscr{E}_{\text{image}}$, mask decoder $\mathscr{D}_{\text{mask}}$, denoiser f_θ
Input: Diffusion steps T, noise schedule $\{\bar{\alpha}_t\}_{t=1}^{T}$
Output: Predicted segmentation mask $\hat{M}$ and evaluation metrics
Encode conditioning image into latent space;
$i_c \leftarrow \mathscr{E}_{\text{image}}(I)$;
Sample latent noise and select final diffusion timestep;
$m_T \sim \mathcal{N}(0, I)$; $\quad t \leftarrow T$;
Predict noise in a single denoising step;
$\hat{\varepsilon} \leftarrow f_\theta(m_T, t; i_c)$;
Estimate clean latent representation of the mask;
$\hat{m}_0 \leftarrow \frac{1}{\sqrt{\bar{\alpha}_t}}(m_T - \sqrt{1-\bar{\alpha}_t}\,\hat{\varepsilon})$;
Decode latent representation into segmentation mask;
$\hat{M}_{\text{soft}} \leftarrow \mathscr{D}_{\text{mask}}(\hat{m}_0)$;
Obtain discrete class predictions;
$\hat{M} \leftarrow \arg\max_c \hat{M}_{\text{soft}}^{(c)}$;
Compute evaluation metrics;
Compute Dice Similarity Coefficient (DSC), IoU, modified IoU (mIoU), 95th percentile Hausdorff distance (HD95), Sensitivity, and Specificity separately for ischemic (IS) and hemorrhagic (HS) classes;
Report lesion-only mean scores as the arithmetic average over {IS, HS}.;

where λ is a weighting coefficient set to 1 in all experiments. This joint optimization enables the denoising U-Net to approximate the underlying noise distribution while also producing structurally consistent latent representations suitable for accurate segmentation.

Reverse process in inference

In conventional diffusion-based generative models, the reverse process is performed over multiple timesteps, beginning with a Gaussian noise sample and gradually denoising it through iterative updates. While such iterative sampling is essential for high-fidelity image synthesis, segmentation tasks typically involve structurally simpler and semantically constrained outputs. Consequently, performing a full reverse diffusion process in segmentation tasks may incur substantial computational cost without yielding significant gains in performance. The inference and evaluation pipeline of the proposed SDMC-Seg framework, including single-step latent reconstruction and metric computation, is summarized in Algorithm 2.

Motivated by the efficient sampling strategy introduced in SDSeg, a single-step latent estimation process is adopted in this work. During inference, the latent representation of the input CT image is obtained *via* the image encoder. A standard Gaussian noise image is sampled and concatenated with the latent embedding of the CT image; the resulting tensor is then passed to the denoising U-Net, which predicts the noise component $\hat{\varepsilon}$. The estimated latent representation of the segmentation map at timestep zero, denoted as $\hat{m}_0$, is obtained using Eq. (12) and subsequently decoded into the final segmentation map by the mask decoder as $S = \mathscr{D}_{\text{mask}}(\hat{m}_0)$. This process requires only a single forward pass through the denoising network, eliminating the need for external samplers or iterative refinement methods such as DDIM. An overview of the inference pipeline is illustrated in Fig. 2. The inclusion of latent reconstruction supervision during training enables the generation of

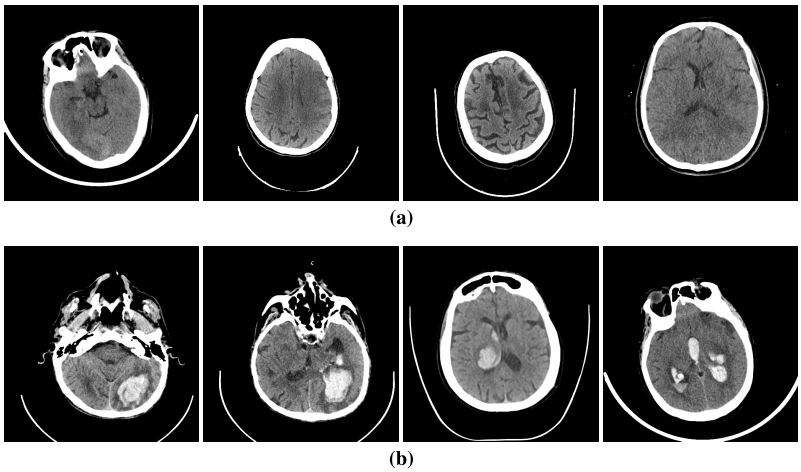


Figure 4 **Sample CT images from the dataset for each case: (A) ischemic, and (B) hemorrhagic stroke.**
Full-size DOI: 10.7717/peerj-cs.3823/fig-4

structurally consistent segmentation outputs in a single denoising step, offering a favorable trade-off between computational efficiency and segmentation accuracy.

EXPERIMENTS

Dataset

This study utilizes a publicly available clinical brain CT dataset provided by the Ministry of Health of the Republic of Türkiye, as described in *Koç et al. (2022)*. The dataset is initially compiled for the TEKNOFEST-2021 Artificial Intelligence in Healthcare competition and includes CT scans collected from various hospitals between 2019 and 2020. Annotations and segmentation masks are created and validated by a team of seven expert radiologists over a six-week period. The final dataset comprises a total of 6,650 axial CT slices, of which 4,427 correspond to patients without any signs of stroke, while the remaining 2,223 images contain confirmed stroke pathology. Among the stroke-positive cases, 1,130 are labeled as ischemic strokes and 1,093 as hemorrhagic strokes, yielding a relatively balanced distribution between the two major subtypes. All images are provided in both DICOM and PNG formats at an original resolution of 512×512 pixels. Each case includes three types of files: the raw CT image, a binary mask image that delineates the stroke region, and an overlay image that visualizes the mask superimposed on the original scan. This comprehensive structure supports both intensity-based analysis and supervised training of segmentation models. The imaging data is curated using structured queries and filtering operations applied to the national e-Pulse and Teleradiology systems, which ensure accurate case selection based on ICD-10 diagnostic codes, radiological procedure codes, and free-text reports. For ischemic stroke cases, diffusion-weighted MRI (DWI) is used as a reference modality during the annotation process to verify subtle or early ischemic findings. In many instances, DWI scans acquired within 1–2 h of the corresponding CT scans are consulted by radiologists to ensure high-quality and clinically consistent labeling of ischemic lesions. Sample CT slices from the dataset are presented in Fig. 4, showing

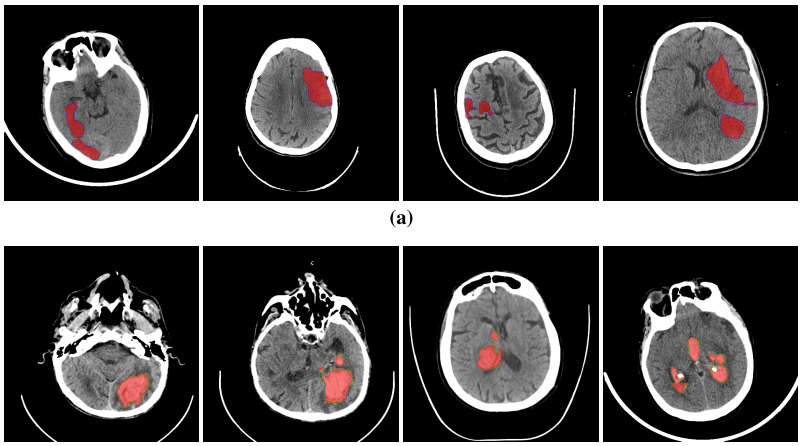


Figure 5 Sample overlay images of (A) ischemic and (B) hemorrhagic stroke cases, as provided by the dataset. Original CT images are blended with expert-annotated masks using semi-transparent coloring; ischemic and hemorrhagic regions are outlined in blue and green, respectively.
Full-size DOI: 10.7717/peerj-cs.3823/fig-5

examples from each case: ischemic, and hemorrhagic. Figure 5 shows the corresponding overlay images for the ischemic and hemorrhagic cases, with expert-annotated lesion masks superimposed on the original CT scans. While the full dataset is described here to provide an overview of its composition, model training and evaluation in this study are conducted using a stroke-positive subset only, as detailed below.

This dataset is utilized for its clinical reliability and expert-labeled segmentations, with particular emphasis on its multi-class structure that enables explicit differentiation between ischemic and hemorrhagic stroke types. In this study, only two-dimensional (2D) CT slices containing confirmed stroke pathology are used; slices without any stroke findings (non-stroke slices) are excluded. This design choice is motivated by the need to prevent the dominance of background-only samples during training and to explicitly focus the learning process on lesion-specific patterns that discriminate between ischemic and hemorrhagic stroke types, which constitute the primary objective of this work. Although non-stroke regions appear as a separate channel in the segmentation masks for evaluation purposes, they represent background tissue rather than an explicit diagnostic class, and the learning objective focuses on discriminating ischemic and hemorrhagic lesion regions. All images and their corresponding segmentation masks are resized to 256×256 pixels to reduce computational cost and ensure compatibility with GPU memory constraints during model training. Although the dataset includes both DICOM and PNG formats, training is performed using the PNG format, which preserves the clinically relevant windowing adjustments selected by radiologists during CT acquisition. This choice ensures that the visual characteristics seen during model training closely reflect the clinical interpretation of stroke findings. To obtain a more robust estimate of generalization and to reduce sensitivity to a single split, a two-stage evaluation protocol is employed. First, 10% of the stroke-positive slices are set aside as an independent, held-out test set using stratified

sampling to preserve the ischemic/hemorrhagic distribution. The remaining 90% of the data are then evaluated using five-fold cross-validation, where in each fold four partitions are used for training and the remaining partition is used for validation. Importantly, the held-out test set is never used during training or validation in any fold. All splits are performed at the slice level due to the absence of patient-level identifiers in the publicly released dataset, which precludes subject-wise partitioning. To enhance robustness and mitigate overfitting, data augmentation is applied to the training set, including random horizontal flipping and random rotation up to ± 15 degrees, each applied with a probability of 20%. These augmentations introduce geometric variability while preserving the anatomical structure of brain tissues. Aside from resizing and augmentation, no additional preprocessing steps (such as intensity normalization or skull stripping) are applied. All preprocessing and dataset preparation procedures are implemented in a fully reproducible manner and are publicly available in the GitHub repository at https://github.com/arozcan/sdmc-seg.

To further mitigate residual class imbalance between ischemic and hemorrhagic stroke samples during training, a class-weighted sampling strategy is employed. Specifically, a weighted random sampler is used to ensure balanced exposure of stroke subtypes across training iterations. The same sampling strategy is consistently applied to all baseline and proposed models to ensure a fair and unbiased comparison.

Implementation details

The overall training pipeline consists of two distinct stages. First, the KL-regularized autoencoder is fine-tuned to encode segmentation masks into a compact latent representation. Then, the conditional denoiser is trained in the latent space to perform multi-class brain stroke segmentation. Both models are trained on an NVIDIA RTX 4060 Ti GPU with 16 GB of memory.

In the first stage, the KL-regularized autoencoder is fine-tuned to encode segmentation masks into a compact latent space. The model follows a convolutional encoder-decoder architecture with channel multipliers of $[1, 2, 4, 4]$, an embedding dimension of 4, and a downsampling factor of 8. The input resolution is set to 256×256, where single-channel segmentation masks are one-hot encoded and expanded to three channels, and the output latent representation has spatial dimensions of 32×32 with four feature channels. The model is trained for 100 epochs using a batch size of 2 and the AdamW optimizer. The training objective combines perceptual Learned Perceptual Image Patch Similarity (LPIPS) loss and adversarial loss, where the discriminator is activated after 50,001 steps. The KL divergence and discriminator losses are weighted by 1×10^{-6} and 0.5, respectively. Only segmentation masks from stroke cases are used during this stage, with preprocessing applied accordingly.

The conditional denoiser is subsequently trained in the latent space to perform multi-class brain stroke segmentation. The model uses a U-Net-based architecture with attention applied at multiple resolutions. It receives eight input channels, resulting from the channel-wise concatenation of the noised latent segmentation and the latent representation of the corresponding CT image, each comprising four channels, and

produces four output channels that reconstruct the segmentation latent. A base channel width of 192 is used, with attention applied at spatial resolutions of 32, 16, 8, and 4. Training is conducted for 300 epochs with a batch size of 4 using the AdamW optimizer. A learning rate scheduler is employed, which linearly increases the learning rate from 1×10^{-6} to 1×10^{-5} during the first 10,000 steps, followed by linear decay. The diffusion process is defined over 1,000 timesteps, where at each training step the noise level t is randomly sampled from the interval $[1, 1{,}000]$, allowing the model to learn denoising across varying levels of noise. The conditional denoiser is initialized with pre-trained weights from the Stable Diffusion model trained on the LSUN-Churches (*Yu et al., 2015*) dataset. The learning rate range and scheduling strategy are informed by established latent diffusion training protocols, including standard Stable Diffusion configurations, and refined through preliminary pilot experiments conducted on the training set. These experiments aim to ensure stable convergence and robust optimization behavior rather than exhaustive hyperparameter tuning. The selected schedule is then fixed and used consistently for all folds and comparative models.

To ensure a consistent and fair comparison, a comprehensive set of baseline segmentation models was implemented and evaluated under the same dataset, preprocessing pipeline, and experimental setup as the proposed SDMC-Seg framework. Specifically, convolutional architectures including U-Net (*Ronneberger, Fischer & Brox, 2015*), ResUNet (*Zhang, Liu & Wang, 2018*), SegResNet (*Myronenko, 2018*), and Attention U-Net (*Oktay et al., 2018*) were considered, alongside transformer-based and hybrid models such as Swin-UNETR (*Hatamizadeh et al., 2021*) and TransUNet (*Chen et al., 2021*). In addition, the self-configuring nnU-Net v2 framework (*Isensee et al., 2021*) was employed as a strong and widely accepted reference baseline.

All baseline segmentation models are trained using the Dice + Cross-Entropy (DiceCE) loss to ensure stable optimization under class imbalance and to maintain consistency across architectures. For convolutional U-Net–based models (U-Net, ResUNet, SegResNet, and Attention U-Net), instance or group normalization and ReLU-based activation functions are employed, without dropout. For transformer-based and hybrid architectures, including TransUNet and Swin-UNETR, the native normalization schemes (*e.g.*, Layer Normalization) and activation mechanisms defined in the original architectures are preserved.

The U-Net architecture consists of five encoding and decoding stages with feature channels configured as $32, 32, 64, 128, 256, 32$ and employs LeakyReLU activation. ResUNet is built upon a ResNet-34 backbone as the encoder, with decoder channels set to $256, 128, 64, 32, 16$ and non-trainable interpolation-based upsampling. SegResNet starts with 24 initial filters, doubling the number of feature channels at each downsampling stage, and employs group normalization throughout the network. Attention U-Net follows the standard attention-gated U-Net formulation, using channel widths of $16, 32, 64, 128, 256$ with corresponding downsampling strides, enabling the suppression of irrelevant background features and enhanced focus on lesion-relevant spatial regions.

For transformer-based baselines, Swin-UNETR utilizes a hierarchical Vision Transformer encoder with a feature size of 96, depth configuration $2, 2, 2, 2$, and

multi-head self-attention settings of $3, 6, 12, 24$ across four stages. TransUNet is implemented using the $R50 - ViT - B_16$ variant, which combines a ResNet-50 convolutional encoder with a Vision Transformer operating on 16×16 patches, enabling joint modeling of local texture information and long-range contextual dependencies. The model is configured for a 256×256 input resolution with identical input and output channel settings as the other baselines.

The nnU-Net v2 framework is employed in its fully automatic configuration mode, without manual architectural or hyperparameter tuning. Network topology, preprocessing steps, data augmentation policies, and training schedules are automatically determined by the nnU-Net pipeline based on dataset characteristics, following the official implementation guidelines.

All baseline models, except nnU-Net v2, are implemented using the MONAI (*Cardoso et al., 2022*) framework and trained for 300 epochs using the AdamW optimizer with a fixed learning rate of 1.6×10^{-4} and a batch size of 16. The same data augmentation strategies, training–validation splits, held-out test set, and 5-fold cross-validation protocol are applied across all models to ensure methodological consistency and a fair comparison with SDMC-Seg.

Evaluation methodology

To facilitate a standardized and fair comparison with the proposed SDMC-Seg framework, a diverse set of representative baseline segmentation models was evaluated under an identical experimental protocol. Specifically, convolutional architectures including U-Net, ResUNet (with a ResNet-34 encoder), SegResNet, and Attention U-Net were considered, alongside transformer-based and hybrid models such as Swin-UNETR and TransUNet. In addition, the self-configuring nnU-Net v2 framework was employed as a strong and widely accepted reference baseline. All baseline models were trained using the same data preprocessing steps, augmentation strategies, training–validation splits, and evaluation pipeline to ensure methodological consistency and a fair comparison across architectures.

To assess segmentation performance, the Dice Similarity Coefficient (DSC) and Intersection over Union (IoU) were employed as the primary quantitative metrics. These metrics are widely accepted in medical image segmentation tasks due to their sensitivity to region overlap and their robustness in evaluating spatial agreement between predictions and ground truth annotations.

The Dice coefficient quantifies the similarity between the predicted segmentation and the ground truth mask, defined as:

$$\mathrm{DSC} = \frac{2\left|M_{\mathrm{pred}} \cap M_{\mathrm{gt}}\right|}{\left|M_{\mathrm{pred}}\right| + \left|M_{\mathrm{gt}}\right|} \tag{14}$$

where M_{pred} and M_{gt} represent the predicted and ground truth pixel sets, respectively. The IoU is computed as:

$$\mathrm{IoU} = \frac{\left|M_{\mathrm{pred}} \cap M_{\mathrm{gt}}\right|}{\left|M_{\mathrm{pred}} \cup M_{\mathrm{gt}}\right|}. \tag{15}$$

To provide a more clinically informative assessment, particularly for small and subtle ischemic lesions, additional metrics are reported beyond overlap-based scores. Specifically, pixel-wise Sensitivity (recall) and Specificity are computed to quantify the model's ability to detect lesion pixels while avoiding false positives, respectively. Moreover, boundary accuracy is evaluated using the 95th percentile Hausdorff Distance (HD95), which is more robust to outliers than the maximum Hausdorff distance and is widely used in medical segmentation benchmarks.

For each class c, Sensitivity and Specificity are defined using true positives (TP_c), false positives (FP_c), false negatives (FN_c), and true negatives (TN_c) computed from the one-hot encoded predictions and ground truth masks:

$$\text{Sensitivity}_c = \frac{TP_c}{TP_c + FN_c}, \tag{16}$$

$$\text{Specificity}_c = \frac{TN_c}{TN_c + FP_c}. \tag{17}$$

HD95 is computed as the 95th percentile of the bidirectional surface-to-surface distances between predicted and ground-truth lesion boundaries for each stroke class, and reported in pixels (or millimeters when spacing information is available). In line with the main analysis, lesion-focused summary scores are additionally reported by aggregating metrics over ischemic and hemorrhagic classes (excluding the non-stroke background) to avoid inflation by the dominant background region.

Final segmentation performance is reported using DSC, IoU, and modified IoU (mIoU), complemented with Sensitivity, Specificity, and HD95 to better reflect lesion detectability and boundary quality. For cross-validation, metrics are computed independently for each fold and summarized as the mean and standard deviation across the five folds, providing a statistically more robust performance estimate. Final evaluation on the held-out test set is conducted using the model selected based on cross-validation performance; soft model outputs are converted to class predictions using softmax and argmax, and subsequently transformed into one-hot encoded masks for metric computation.

In this study, an additional evaluation metric termed modified IoU (mIoU), a variant of the standard IoU, is incorporated to align with the evaluation protocol introduced in the dataset publication (*Koç et al., 2022*). mIoU accounts for minor boundary inconsistencies that may arise due to inter-observer variability, even among expert radiologists. Specifically, it is calculated as the intersection between the predicted mask and the ground truth mask dilated with a 3×3 kernel, divided by the union between the predicted mask and the ground truth mask eroded with the same kernel. The metric is formally defined as:

$$\text{mIoU} = \frac{\left|M_{\text{pred}} \cap \text{dilate}(M_{\text{gt}})\right|}{\left|M_{\text{pred}} \cup \text{erode}(M_{\text{gt}})\right|}, \tag{18}$$

where operators $\text{dilate}(\cdot)$ and $\text{erode}(\cdot)$ represent morphological dilation and erosion using a 3×3 kernel. Since M_{gt} contains multiple semantic classes, each class is processed independently during dilation and erosion, and the results are merged prior to computing the intersection and union.

Table 1 Autoencoder reconstruction metrics for standard and fine-tuned models. Metrics are reported across all classes and excluding the non-stroke (NS) background class.

Model	DSC	DSC (w/o NS)	IoU	IoU (w/o NS)
Standard	0.9621	0.9247	0.9373	0.8755
Fine-tuned	0.9972	0.9945	0.9945	0.9891

Lastly, to isolate the effect of key architectural choices introduced in SDMC-Seg, an ablation study was conducted. This analysis evaluates the individual contributions of the fine-tuned mask encoder and the trainable image encoder components. Results and qualitative comparisons from these ablation experiments are provided in the Results section.

RESULTS

Autoencoder reconstruction performance

The reconstruction quality of the KL-regularized autoencoder is evaluated in two configurations: (i) using the original pre-trained weights from Stable Diffusion (referred to as the standard model), and (ii) after the model is fine-tuned on stroke segmentation masks (fine-tuned model).

Reconstruction accuracy is measured using the DSC and IoU, both computed by excluding the non-stroke (NS) regions. As shown in Table 1, the standard model yields a DSC of 0.9247 and an IoU of 0.8755, while the fine-tuned model achieves 0.9945 and 0.9891, respectively. These results indicate that fine-tuning substantially improves the reconstruction quality of stroke regions that are clinically relevant.

These results indicate that fine-tuning on domain-specific segmentation masks substantially improves the model's ability to reconstruct clinically relevant stroke regions with high structural fidelity. Qualitative comparisons of reconstructed masks and their corresponding latent representations are provided in Fig. 3.

Segmentation performance

Segmentation performance is evaluated using the DSC, IoU, modified IoU (mIoU), and boundary-based metrics. All metrics are computed separately for ischemic (IS) and hemorrhagic (HS) lesion classes to enable class-wise assessment under severe class imbalance. Since the non-stroke (NS) background class dominates the pixel distribution and can artificially inflate overlap-based scores, lesion-only mean values are reported as the arithmetic average of IS and HS scores, excluding the NS class.

Table 2 reports the five-fold cross-validation results in terms of DSC and IoU. Among all evaluated models, the proposed SDMC-Seg consistently achieves the highest lesion-only mean performance, attaining a mean DSC of 0.745 and a mean IoU of 0.656 across ischemic and hemorrhagic regions. These results suggest consistently stronger overlap performance relative to both convolutional and transformer-based baselines. In particular, SDMC-Seg demonstrates improved ischemic lesion segmentation, which remains a challenging task due to the subtle and low-contrast appearance of early ischemic

Table 2 Comparison of five-fold cross-validation segmentation performance across models in terms of class-wise and lesion-only mean DSC and IoU scores. IS and HS denote ischemic and hemorrhagic stroke regions, respectively. Mean scores are computed as the arithmetic average of the IS and HS classes, excluding the non-stroke (NS) background. Reported values are given as mean ± standard deviation across folds.

Model	DSC			IoU		
	IS	HS	Mean	IS	HS	Mean
U-Net	0.620 ± 0.022	0.798 ± 0.026	0.709 ± 0.016	0.508 ± 0.020	0.707 ± 0.025	0.607 ± 0.014
ResUNet (R34)	0.629 ± 0.019	0.793 ± 0.029	0.711 ± 0.021	0.517 ± 0.019	0.704 ± 0.028	0.610 ± 0.020
SegResNet	0.620 ± 0.032	0.807 ± 0.025	0.714 ± 0.020	0.511 ± 0.030	0.717 ± 0.030	0.614 ± 0.021
Swin-UNETR	0.605 ± 0.020	0.794 ± 0.033	0.699 ± 0.014	0.496 ± 0.021	0.703 ± 0.034	0.599 ± 0.015
Attention U-Net	0.587 ± 0.030	0.779 ± 0.029	0.683 ± 0.015	0.475 ± 0.027	0.686 ± 0.029	0.581 ± 0.014
TransUNet	0.464 ± 0.017	0.740 ± 0.044	0.602 ± 0.022	0.363 ± 0.014	0.641 ± 0.044	0.502 ± 0.022
nnU-Net v2	0.626 ± 0.013	0.813 ± 0.029	0.718 ± 0.019	0.531 ± 0.014	0.730 ± 0.030	0.629 ± 0.018
SDMC-Seg (Proposed)	0.668 ± 0.038	0.822 ± 0.020	0.745 ± 0.019	0.575 ± 0.041	0.737 ± 0.020	0.656 ± 0.024

changes in non-contrast CT images. Detailed fold-wise DSC and IoU results for all models are provided in Supplemental Section S1.

To further assess spatial robustness and boundary alignment, Table 3 summarizes the five-fold cross-validation results in terms of mIoU and the 95th percentile Hausdorff distance (HD95). Among all evaluated methods, SDMC-Seg achieves the highest lesion-only mean mIoU (0.778) while maintaining competitive HD95 values, indicating improved boundary consistency alongside strong region-level agreement. This improvement is particularly notable for ischemic regions, which are known to exhibit subtle and low-contrast appearances in non-contrast CT imaging. Detailed fold-wise mIoU and HD95 results for all models are provided in Supplemental Section S1.

In addition to overlap and boundary-based metrics, clinical relevance is evaluated using sensitivity and specificity, as summarized in Table 4. The proposed method yields the highest lesion detection sensitivity among all models while maintaining consistently high specificity. This balance indicates enhanced sensitivity to clinically relevant pathological patterns without an accompanying increase in false-positive detections, which is particularly important for reliable clinical decision-support systems. The uniformly high specificity across models is largely due to the dominance of background pixels, which can inflate true-negative rates in highly imbalanced segmentation tasks. Fold-wise sensitivity and specificity values for each model are summarized in Supplemental Section S1.

All reported results correspond to the mean performance obtained under five-fold cross-validation on the training and validation subset, comprising 90% of the dataset. Fold-wise performance values underlying these summaries are reported in Supplemental Section S1. The held-out test set is excluded from cross-validation and is reserved exclusively for final generalization assessment, with corresponding results presented in Supplemental Section S2.

Qualitative segmentation results for representative stroke cases are presented in Fig. 6. The figure compares the predicted segmentation outputs of the proposed SDMC-Seg model with those produced by conventional convolutional and transformer-based baselines. Visual inspection reveals that SDMC-Seg consistently produces segmentation

Table 3 Five-fold cross-validation results in terms of mIoU and HD95. mIoU accounts for boundary tolerance as defined in the dataset protocol, while HD95 evaluates boundary alignment accuracy. Lower HD95 values indicate better segmentation quality. IS and HS denote ischemic and hemorrhagic stroke lesions, respectively. Mean values are computed as the arithmetic average of IS and HS, excluding the non-stroke (NS) class.

Model	mIoU			HD95 (↓)		
	IS	HS	Mean	IS	HS	Mean
U-Net	0.598 ± 0.023	0.846 ± 0.029	0.722 ± 0.017	26.54 ± 1.28	13.14 ± 2.05	20.04 ± 0.55
ResUNet (R34)	0.611 ± 0.022	0.844 ± 0.032	0.727 ± 0.020	22.17 ± 2.30	12.37 ± 1.46	17.41 ± 0.89
SegResNet	0.601 ± 0.035	0.859 ± 0.030	0.730 ± 0.021	26.90 ± 1.26	13.08 ± 3.03	20.09 ± 2.15
Swin-UNETR	0.586 ± 0.023	0.843 ± 0.033	0.715 ± 0.015	24.46 ± 1.55	13.52 ± 1.19	19.11 ± 1.08
Attention U-Net	0.562 ± 0.031	0.827 ± 0.032	0.694 ± 0.016	29.96 ± 1.91	14.07 ± 2.00	22.22 ± 1.23
TransUNet	0.428 ± 0.017	0.787 ± 0.048	0.608 ± 0.024	33.50 ± 1.31	16.16 ± 2.71	24.95 ± 1.81
nnU-Net v2	0.620 ± 0.014	0.869 ± 0.031	0.743 ± 0.020	16.87 ± 1.40	8.69 ± 1.63	12.77 ± 0.37
SDMC-Seg (Proposed)	0.670 ± 0.045	0.887 ± 0.025	0.778 ± 0.020	16.30 ± 2.78	7.71 ± 2.41	13.00 ± 2.25

Table 4 Five-fold cross-validation performance comparison across models in terms of sensitivity and specificity. IS and HS denote ischemic and hemorrhagic stroke lesions, respectively. Mean values are computed as the arithmetic average of IS and HS, excluding the non-stroke (NS) background.

Model	Sensitivity			Specificity		
	IS	HS	Mean	IS	HS	Mean
U-Net	0.648 ± 0.019	0.803 ± 0.020	0.724 ± 0.017	0.995 ± 0.001	0.999 ± 0.000	0.997 ± 0.000
ResUNet (R34)	0.649 ± 0.029	0.786 ± 0.033	0.717 ± 0.025	0.995 ± 0.002	0.999 ± 0.000	0.997 ± 0.001
SegResNet	0.647 ± 0.031	0.819 ± 0.020	0.731 ± 0.017	0.995 ± 0.000	0.999 ± 0.000	0.997 ± 0.000
Swin-UNETR	0.605 ± 0.021	0.797 ± 0.040	0.699 ± 0.014	0.996 ± 0.000	0.999 ± 0.000	0.998 ± 0.000
Attention U-Net	0.595 ± 0.038	0.782 ± 0.037	0.688 ± 0.019	0.996 ± 0.001	0.999 ± 0.000	0.997 ± 0.000
TransUNet	0.466 ± 0.021	0.737 ± 0.042	0.599 ± 0.025	0.994 ± 0.000	0.999 ± 0.000	0.996 ± 0.000
nnU-Net v2	0.630 ± 0.015	0.801 ± 0.027	0.714 ± 0.019	0.997 ± 0.000	0.999 ± 0.000	0.998 ± 0.000
SDMC-Seg (Proposed)	0.667 ± 0.052	0.823 ± 0.031	0.743 ± 0.029	0.997 ± 0.001	0.999 ± 0.000	0.998 ± 0.001

masks that exhibit improved spatial coherence, sharper lesion boundaries, particularly in challenging ischemic cases.

For hemorrhagic stroke regions, where lesion boundaries are generally more distinct, most models achieve reasonable localization; however, SDMC-Seg demonstrates improved boundary adherence and fewer spurious activations in surrounding healthy tissue. In ischemic stroke cases, which are characterized by subtle intensity variations and poorly defined borders in non-contrast CT images, baseline models frequently produce fragmented or incomplete segmentations. In contrast, SDMC-Seg yields more contiguous and anatomically plausible lesion regions, indicating enhanced sensitivity to weak pathological cues.

These qualitative improvements align with the quantitative gains observed in DSC, IoU, and mIoU metrics, and highlight the practical advantages of the proposed latent diffusion-based formulation. In particular, the ability of SDMC-Seg to suppress false positives while preserving lesion continuity suggests improved robustness under severe class imbalance, reinforcing its suitability for clinically relevant stroke segmentation tasks.

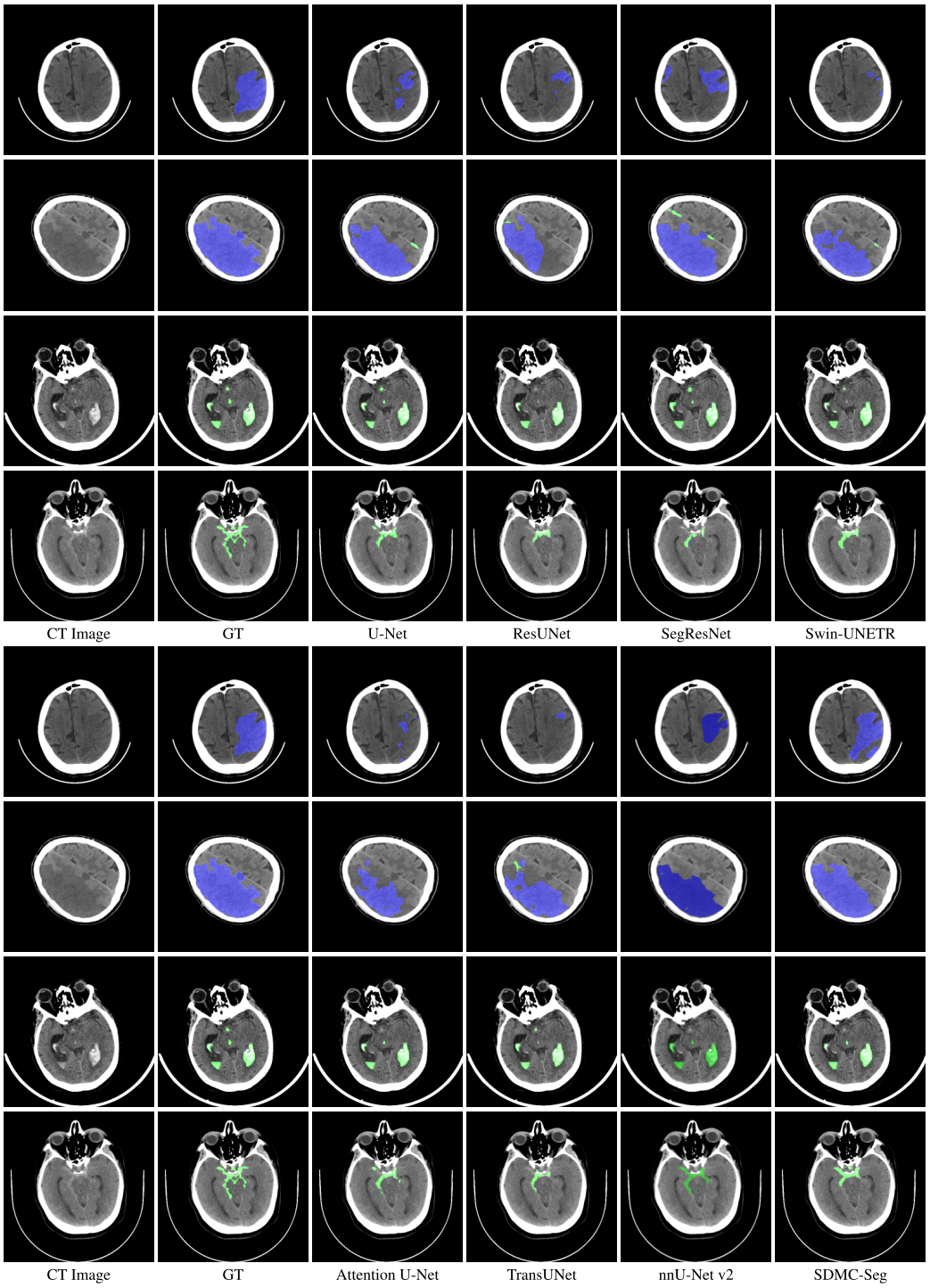


Figure 6 Qualitative segmentation results for representative stroke cases across all evaluated models. Each example includes the input CT image, the corresponding ground truth annotation, and predicted segmentation outputs generated by U-Net, ResUNet, SegResNet, Attention U-Net, TransUNet, Swin-UNETR, nnU-Net v2, and the proposed SDMC-Seg. Hemorrhagic and ischemic lesion regions are visualized in green and blue, respectively. Full-size DOI: 10.7717/peerj-cs.3823/fig-6

Table 5 Ablation study of SDMC-Seg under different encoder configurations. Reported values are lesion-only mean scores computed as the arithmetic average over ischemic and hemorrhagic classes, excluding the non-stroke (NS) background. In addition to overlap-based metrics (DSC, IoU, mIoU), boundary accuracy (HD95) and clinical detection metrics (Sensitivity, Specificity) are reported.

Mask/Image encoder	DSC	IoU	mIoU	HD95 ↓	Sensitivity	Specificity
Pretrained/Trainable	0.634 ± 0.023	0.529 ± 0.021	0.653 ± 0.023	19.17 ± 2.03	0.646 ± 0.032	0.998 ± 0.001
Fine-tuned/Frozen	0.507 ± 0.028	0.455 ± 0.027	0.521 ± 0.029	22.57 ± 2.49	0.517 ± 0.035	0.996 ± 0.001
Fine-tuned/Trainable	0.745 ± 0.019	0.656 ± 0.024	0.778 ± 0.020	13.00 ± 2.25	0.743 ± 0.029	0.998 ± 0.001

Ablation study

To evaluate the contributions of encoder configurations in SDMC-Seg, a series of ablation experiments is conducted focusing on two key modifications introduced over the standard Stable Diffusion design: fine-tuning the mask encoder and allowing the conditional image encoder to be trainable. Three configurations are examined: (i) pretrained mask encoder with a trainable image encoder, (ii) fine-tuned mask encoder with a frozen image encoder, and (iii) fine-tuned mask encoder with a trainable image encoder, as summarized in Table 5. This experimental setup enables a systematic assessment of how each design choice influences segmentation accuracy.

The effect of fine-tuning the mask encoder is first analyzed by comparing models initialized with pretrained and fine-tuned weights, while keeping the image encoder trainable in both cases. The results show that the fine-tuned configuration yields consistently higher segmentation accuracy across all reported metrics. This improvement is primarily attributed to enhanced reconstruction performance of the mask autoencoder, which produces more compact and semantically structured latent representations. These representations facilitate more accurate noise estimation during the diffusion process, ultimately resulting in segmentation outputs that better align with the ground truth. Representative segmentation outputs from both configurations are presented in Fig. 7, qualitatively illustrating the improved spatial alignment achieved when using fine-tuned mask encoder weights.

To further assess the role of the image encoder, a comparison is made between models with frozen and trainable image encoders, both using fine-tuned mask encoder weights. A substantial performance drop is observed when the image encoder is kept frozen, despite using the same optimized latent representation. This result highlights the critical importance of adapting the image encoder to the target domain: a static encoder fails to capture relevant domain-specific features, thereby weakening the effectiveness of the conditioning pathway. In contrast, a trainable image encoder enables the model to extract richer contextual information from CT images, considerably enhanced segmentation accuracy. These findings collectively confirm that jointly optimizing both encoder components is essential for effective conditional latent diffusion–based segmentation in the proposed framework.

DISCUSSION

The results presented in this study demonstrate the effectiveness of SDMC-Seg, a latent diffusion-based model tailored for multi-class brain stroke segmentation in CT images.

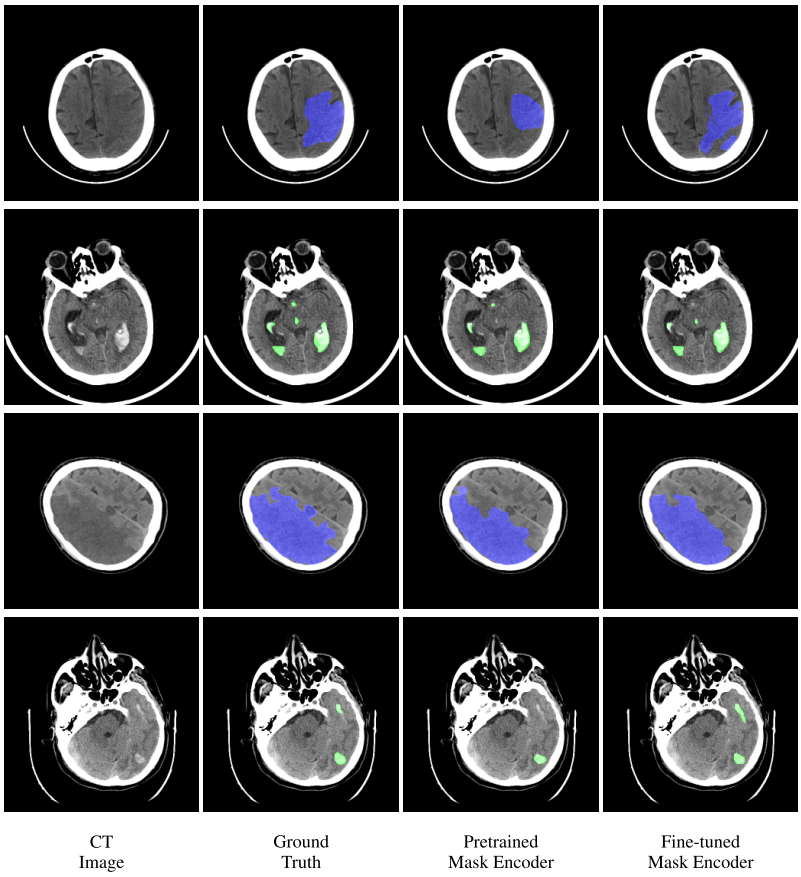


Figure 7 Qualitative comparison of segmentation results produced by SDMC-Seg models initialized with pretrained and fine-tuned mask encoder weights. Full-size DOI: 10.7717/peerj-cs.3823/fig-7

Unlike conventional segmentation architectures, SDMC-Seg combines a denoising diffusion model with compact latent representations learned through a mask autoencoder. This design leads to improved accuracy, particularly in ischemic and hemorrhagic regions, which are clinically critical yet challenging to delineate.

A key factor contributing to the performance of SDMC-Seg lies in its architectural choices, particularly the design and optimization of its encoder components. Fine-tuning the mask encoder enhances the quality of the latent space, enabling the learning of representations that are both compact and semantically structured. These representations facilitate more accurate noise estimation during the diffusion process. Furthermore, allowing the image encoder to be trainable enables the conditioning pathway to adapt to the target domain, thereby capturing stroke-relevant features that static encoders may overlook. Ablation studies further confirm that jointly adapting both encoder components leads to substantial gains in segmentation accuracy.

Compared to established baselines such as U-Net, ResUNet, SegResNet, Attention U-Net, TransUNet, Swin-UNETR, and nnU-Net v2, SDMC-Seg demonstrates consistently strong performance, particularly in stroke-specific classes. While baseline models achieve comparable accuracy in hemorrhagic regions, reflected by uniformly high specificity

scores, their sensitivity to ischemic lesions remains limited. This behavior is expected given the overwhelming dominance of non-stroke background pixels, where false-positive suppression is comparatively easier for all models. In contrast, the proposed SDMC-Seg achieves higher lesion-wise DSC, IoU, and sensitivity for ischemic regions, as shown in Tables 2 and 4. This improvement can be attributed to the latent diffusion formulation combined with a class-wise segmentation design, which implicitly captures uncertainty in lesion boundaries and enables progressive refinement of ambiguous regions during the denoising process. As a result, SDMC-Seg exhibits enhanced robustness in cases where ischemic lesions present minimal contrast and poorly defined boundaries in non-contrast CT images.

Beyond baseline comparisons, several previous works are examined to provide additional context for evaluating the effectiveness of SDMC-Seg. Among these, *Koç et al. (2022)* introduce the dataset through an artificial intelligence competition, where segmentation performance is evaluated in the second stage by tasking participating teams with delineating ischemic and hemorrhagic regions on non-contrast CT images. However, segmentation performance in this competition is assessed using a merged evaluation strategy without class-wise separation, reporting a single IoU score that combines ischemic and hemorrhagic regions. Moreover, instead of employing the standard IoU definition, a modified evaluation metric is used, which accounts for minor boundary inconsistencies by applying morphological dilation and erosion operations to the ground truth masks before intersection and union computation. According to the results presented by *Koç et al. (2022)*, a mean modified IoU score of 0.785 is achieved by the top three teams. Due to the unavailability of ground truth segmentation masks for Stage 2, it is not possible to perform a direct evaluation or comparative analysis of segmentation results on this subset.

In addition to CT-based approaches, *Zhang, Confidence & Anazodo (2022)* proposed a similarity-weighted self-ensembling framework for stroke lesion segmentation under low-quality and few-shot MRI settings. Their method focuses on mitigating annotation scarcity and image degradation by leveraging self-ensembling strategies and similarity-based weighting across limited training samples. While this approach demonstrates promising results in MRI-based stroke segmentation, it relies on multi-modal and higher-contrast imaging, as well as few-shot learning assumptions that differ substantially from the acute non-contrast CT scenario addressed in this study. In contrast, SDMC-Seg is designed to operate under standard clinical CT conditions without relying on few-shot supervision or multi-modal inputs, emphasizing robustness and efficiency in emergency settings.

In a related study, *Yalçın & Vural (2022)* proposed a modified U-Net architecture enhanced with a dimensional transform block, enabling the network to jointly perform brain stroke classification and segmentation tasks using the same CT dataset. Segmentation performance is evaluated using a modified IoU metric, applying morphological dilation and erosion to ground truth masks, and averaging IoU scores across individual images. In their methodology, the number of stroke images is artificially increased by duplicating samples to balance the class distribution. Since data augmentation and duplication are performed before the random train-test split, similar or augmented

versions of the same images could potentially appear in both sets, and their evaluation metric may have influenced the observed performance. Due to the differences in data preparation strategies, including augmentation and train-test splitting methodologies, a direct performance comparison cannot be reliably interpreted.

Furthermore, a comparative analysis of single-stage and hierarchical approaches for multi-type stroke lesion segmentation is presented by *Samak (2025)* using the same CT dataset. Segmentation performance is reported separately for cases that include and exclude non-stroke (healthy) slices. In this evaluation, the highest segmentation scores are obtained using conventional architectures such as U-Net and DeepLabV3, with metrics computed as averages across all segmentation classes, including the non-stroke background. To ensure comparability under analogous evaluation settings, corresponding results are computed in this study by including the non-stroke background class in the metric calculations. Under these conditions, the proposed SDMC-Seg model achieves higher segmentation scores than those reported for U-Net and DeepLabV3 in *Samak (2025)*, demonstrating its effectiveness under the same evaluation criteria.

Taken together, these studies highlight that reported segmentation performance is strongly influenced by evaluation protocols, metric definitions, data preparation strategies, and imaging modalities. In contrast to approaches relying on modified metrics, data duplication, or multi-modal inputs, SDMC-Seg is evaluated under a strictly lesion-aware and class-separated protocol using standard non-contrast CT images. This positioning emphasizes robustness and clinical realism rather than metric-driven optimization.

More recently, *Di Matteo et al. (2025)* investigated deep learning approaches for segmenting sub-acute and chronic stroke lesions using multi-modal MRI data. Although multi-modal inputs can substantially enhance lesion visibility and segmentation accuracy, such approaches depend on the availability of multiple imaging modalities. This requirement may limit their applicability in acute or emergency settings, where non-contrast CT remains the primary and most accessible imaging modality.

By leveraging a single-step reverse diffusion process, SDMC-Seg avoids the iterative denoising typically required in diffusion-based segmentation models. This design leads to a streamlined inference procedure that reduces computational complexity while preserving structural fidelity in the predicted segmentation masks. Although explicit runtime benchmarking is beyond the scope of this study, the single-step formulation offers a practical advantage over multi-step diffusion-based approaches in time-constrained clinical settings.

A limitation of this study is that model evaluation is conducted at the slice level rather than the patient level. This is primarily due to the anonymized nature of the publicly released dataset, which does not provide patient identifiers or scan-level grouping information required for subject-wise aggregation. While slice-level evaluation is appropriate for assessing lesion delineation accuracy and is consistent with common practice in stroke segmentation studies, patient-level assessment would be more directly aligned with clinical decision-making. Future studies will explore patient-level evaluation using datasets that include subject-wise metadata and volumetric scan information. Another limitation is that the current study focuses on 2D slice-wise segmentation, which

does not explicitly model inter-slice continuity. Although this design choice aligns with the available annotations and common practice, extending the framework to volumetric 3D diffusion-based segmentation remains an important direction for future work.

It is important to clarify that the primary contribution of this work does not lie in proposing a fundamentally new diffusion formulation, but in adapting and systematically validating latent diffusion principles for a clinically relevant multi-class brain stroke segmentation problem. While many existing diffusion-based segmentation methods focus on binary settings or rely on multi-step reverse diffusion procedures, SDMC-Seg demonstrates that single-step latent diffusion can be effectively stabilized and applied to multi-class brain stroke segmentation by jointly modeling ischemic and hemorrhagic lesions as separate semantic channels in a shared latent space. This provides empirical evidence that latent diffusion models can be reliably extended to clinically realistic, domain-specific multi-class segmentation tasks in non-contrast CT imaging.

Despite the observed improvements in quantitative metrics, a more fine-grained failure case analysis, such as stratification based on infarct size or time-to-onset, could not be conducted in this study. While the utilized public dataset includes scans collected from multiple centers, it does not provide reliable patient-level clinical metadata, including stroke onset timing, volumetric lesion measurements, or scan-level grouping information. Consequently, quantitative subgroup analyses targeting early ischemic changes remain beyond the scope of the present work. Addressing these aspects will require future studies using datasets with richer patient-level annotations and volumetric scan information.

CONCLUSIONS

This study introduces SDMC-Seg, a latent diffusion-based framework specifically designed for multi-class stroke segmentation in non-contrast CT. The architecture integrates a fine-tuned mask autoencoder, a trainable CT image encoder, and a conditional denoising network that operates in latent space. By learning and inferring in this compact representation, SDMC-Seg captures semantic structure more effectively and improves accuracy across both ischemic and hemorrhagic stroke types. The method directly addresses key challenges such as the low contrast of ischemic lesions and the severe class imbalance in clinical data. Notably, the segmentation mask is reconstructed in a single denoising step, bypassing the iterative sampling used in traditional diffusion models. This enables structurally consistent predictions within a simplified inference pipeline, which is particularly relevant for time-sensitive clinical settings.

Extensive experimental results show that SDMC-Seg achieves competitive performance, and in several settings outperforms widely used segmentation models including U-Net, ResUNet, SegResNet, Attention U-Net, TransUNet, Swin-UNETR, and nnU-Net v2 particularly in lesion-focused metrics and ischemic stroke segmentation, where accurate delineation remains most challenging. Ablation studies further confirm that jointly adapting both the image and mask encoders to the target domain leads to consistent improvements across overlap-, boundary-, and sensitivity-based metrics, validating the effectiveness of each architectural component.

In addition to its segmentation performance, SDMC-Seg is optimized for computational efficiency. By learning a direct mapping from noisy latent vectors to segmentation masks, the model avoids the iterative reverse diffusion process commonly used in diffusion-based frameworks, enabling a streamlined one-pass inference procedure.

A key novelty of SDMC-Seg lies in its use of class-aware latent mask encoding, where each stroke subtype is represented by a separate semantic channel during training and reconstruction. This design improves multi-class separation and enhances the model's ability to preserve fine anatomical details in challenging cases.

While the model performs robustly, especially in hemorrhagic stroke cases, it shows reduced sensitivity to early-stage ischemic lesions, which are often ambiguous in non-contrast CT. Future work will explore integrating complementary modalities, such as diffusion-weighted MRI or perfusion CT, to enhance early lesion detection and broaden clinical applicability.

ACKNOWLEDGEMENTS

During the preparation of this work, the author used ChatGPT (OpenAI, GPT-5, December 2025 version) and DeepL Translator (DeepL SE, version 5.4.1) to assist with language editing and to improve the overall readability of the manuscript. No AI-generated content (such as original text, figures, or code) was used in the scientific development or substantive writing of the article. The author has reviewed and edited all content and takes full responsibility for the final version of the manuscript.

ADDITIONAL INFORMATION AND DECLARATIONS

Funding

The authors received no funding for this work.

Competing Interests

The authors declare that they have no competing interests.

Author Contributions

- Ahmet Remzi Özcan conceived and designed the experiments, performed the experiments, analyzed the data, performed the computation work, prepared figures and/or tables, authored or reviewed drafts of the article, and approved the final draft.

Data Availability

The following information was supplied regarding data availability:

The source code, training scripts, preprocessing pipelines, and evaluation utilities used in this study are available at Zenodo: arozcan. (2026). arozcan/sdmc-seg: SDMC-Seg v1.1.0 – Revised version supporting PeerJ resubmission (v1.1). Zenodo. https://doi.org/10.5281/zenodo.18189474.

The repository includes full instructions for reproducing the experiments reported.

The raw CT imaging data analyzed in this study are available at Figshare: Özcan, Ahmet Remzi (2026). Multi-Class Brain Stroke CT Segmentation Dataset (Processed). figshare. Dataset. https://doi.org/10.6084/m9.figshare.31057012.v1.

The dataset contains processed and anonymized non-contrast brain CT images and corresponding multi-class stroke lesion segmentation masks. The data are derived from a publicly available national stroke CT dataset provided through the Open Data Portal of the Ministry of Health of the Republic of Türkiye and were originally released for the TEKNOFEST-2021 Artificial Intelligence in Healthcare competition.

Supplemental Information

Supplemental information for this article can be found online at http://dx.doi.org/10.7717/peerj-cs.3823#supplemental-information.

REFERENCES

Cardoso MJ, Li W, Brown R, Ma N, Kerfoot E, Wang Y, Murrey B, Myronenko A, Zhao C, Yang D, Nath V, He Y, Xu Z, Hatamizadeh A, Zhu W, Liu Y, Zheng M, Tang Y, Yang I, Zephyr M, Hashemian B, Alle S, Zalbagi Darestani M, Budd C, Modat M, Vercauteren T, Wang G, Li Y, Hu Y, Fu Y, Gorman B, Johnson H, Genereaux B, Erdal BS, Gupta V, Diaz-Pinto A, Dourson A, Maier-Hein L, Jaeger PF, Baumgartner M, Kalpathy-Cramer J, Flores M, Kirby J, Cooper LAD, Roth HR, Xu D, Bericat D, Floca R, Zhou SK, Shuaib H, Farahani K, Maier-Hein KH, Aylward S, Dogra P, Ourselin S, Feng A. 2022. MONAI: an open-source framework for deep learning in healthcare. ArXiv DOI 10.48550/arXiv.2211.02701.

Chen J, Lu Y, Yu Q, Luo X, Adeli E, Wang Y, Lu L, Yuille AL, Zhou Y. 2021. TransUNet: transformers make strong encoders for medical image segmentation. ArXiv DOI 10.48550/arXiv.2102.04306.

Chen T, Wang C, Shan H. 2023. BerDiff: conditional Bernoulli diffusion model for medical image segmentation. In: *International Conference on Medical Image Computing and Computer-Assisted Intervention*. Cham: Springer, 491–501.

Dhariwal P, Nichol A. 2021. Diffusion models beat GANs on image synthesis. ArXiv DOI 10.48550/arXiv.2105.05233.

Di Matteo A, Mahé Y, Leplaideur SS, Bonan I, Bannier E, Galassi F. 2025. Deep learning and multi-modal MRI for the segmentation of sub-acute and chronic stroke lesions. *Pattern Recognition Letters* **199**:225–231 DOI 10.1016/j.patrec.2025.11.017.

Feigin VL, Stark BA, Johnson CO, Roth GA, Bisignano C, Abady GG, Abbasifard M, Abbasi-Kangevari M, Abd-Allah F, Abedi V, Abualhasan A, Abu-Rmeileh NME, Abushouk AI, Adebayo OM, Agarwal G, Agasthi P, Ahinkorah BO, Ahmad S, Ahmadi S, Salih YA, Aji B, Akbarpour S, Akinyemi RO, Al Hamad H, Alahdab F, Alif SM, Alipour V, Aljunid SM, Almustanyir S, Al-Raddadi RM, Al-Shahi Salman R, Alvis-Guzman N, Ancuceanu R, Anderlini D, Anderson JA, Ansar A, Antonazzo IC, Arabloo J, ÄrnlÖv J, Artanti KD, Aryan Z, Asgari S, Ashraf T, Athar M, Atreya A, Ausloos M, Baig AA, Baltatu OC, Banach M, Barboza MA, Barker-Collo SL, Bärnighausen TW, Barone MTU, Basu S, Bazmandegan G, Beghi E, Beheshti M, Béjot Y, Bell AW, Bennett DA, Bensenor IM, Bezabhe WM, Bezabih YM, Bhagavathula AS, Bhardwaj P, Bhattacharyya K, Bijani A, Bikbov B, Birhanu MM, Boloor A, Bonny A, Brauer M, Brenner H, Bryazka D, Butt ZA, Murray CJL. 2021. Global, regional, and national burden of stroke and its risk factors,

1990–2019: a systematic analysis for the global burden of disease study 2019. *The Lancet Neurology* **20(10)**:795–820 DOI 10.1016/s1474-4422(21)00252-0.

Frid-Adar M, Diamant I, Klang E, Amitai M, Goldberger J, Greenspan H. 2018. GAN-based synthetic medical image augmentation for increased CNN performance in liver lesion classification. *Neurocomputing* **321(5)**:321–331 DOI 10.1016/j.neucom.2018.09.013.

Guo X, Yang Y, Ye C, Lu S, Peng B, Huang H, Xiang Y, Ma T. 2023. Accelerating diffusion models via pre-segmentation diffusion sampling for medical image segmentation. In: *2023 IEEE 20th International Symposium on Biomedical Imaging (ISBI)*. Piscataway: IEEE, 1–5.

Hatamizadeh A, Nath V, Tang Y, Yang D, Roth HR, Xu D. 2021. Swin UNETR: swin transformers for semantic segmentation of brain tumors in MRI images. In: *International MICCAI Brainlesion Workshop*. Cham: Springer, 272–284.

Ho J, Jain A, Abbeel P. 2020. Star-shaped denoising diffusion probabilistic models. In: *34th Conference on Neural Information Processing Systems (NeurIPS 2020), Vancouver, Canada*.

Hollist M, Morgan L, Cabatbat R, Au K, Kirmani MF, Kirmani BF. 2021. Acute stroke management: overview and recent updates. *Aging and Disease* **12(4)**:1000 DOI 10.14336/ad.2021.0311.

Isensee F, Jaeger PF, Kohl SA, Petersen J, Maier-Hein KH. 2021. nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation. *Nature Methods* **18(2)**:203–211 DOI 10.1038/s41592-020-01008-z.

Kazerouni A, Aghdam EK, Heidari M, Azad R, Fayyaz M, Hacihaliloglu I, Merhof D. 2023. Diffusion models in medical imaging: a comprehensive survey. *Medical Image Analysis* **88(1)**:102846 DOI 10.1016/j.media.2023.102846.

Kingma DP, Welling M. 2019. An introduction to variational autoencoders. *Foundations and Trends in Machine Learning* **12(4)**:307–392 DOI 10.1561/2200000056.

Koç U, Akçapınar Sezer E, Özkaya YA, Yarbay Y, Taydaş O, Ayyıldız VA, Kızıloğlu HA, Kesimal U, Çankaya İ, Beşler MS, Karakaş E, Karademir F, Sebik NB, Bahadır M, Sezer Ö, Yeşilyurt B, Varlı S, Akdoğan E, Ülgü MM, Birinci Ş. 2022. Artificial intelligence in healthcare competition (TEKNOFEST-2021): stroke data set. *The Eurasian Journal of Medicine* **54(3)**:248–258 DOI 10.5152/eurasianjmed.2022.22096.

Kumar A, Ghosal P, Kundu SS, Mukherjee A, Nandi D. 2022. A lightweight asymmetric U-Net framework for acute ischemic stroke lesion segmentation in CT and CTP images. *Computer Methods and Programs in Biomedicine* **226(1)**:107157 DOI 10.1016/j.cmpb.2022.107157.

Kumar A, Upadhyay N, Ghosal P, Chowdhury T, Das D, Mukherjee A, Nandi D. 2020. CSNet: a new DeepNet framework for ischemic stroke lesion segmentation. *Computer Methods and Programs in Biomedicine* **193(1)**:105524 DOI 10.1016/j.cmpb.2020.105524.

Lin T, Chen Z, Yan Z, Yu W, Zheng F. 2024. Stable diffusion segmentation for biomedical images with single-step reverse process. In: *International Conference on Medical Image Computing and Computer-Assisted Intervention*. Cham: Springer, 656–666.

Liu Q, Yu L, Luo L, Dou Q, Heng PA. 2020. Semi-supervised medical image classification with relation-driven self-ensembling model. *IEEE Transactions on Medical Imaging* **39(11)**:3429–3440 DOI 10.1109/tmi.2020.2995518.

Martin SS, Aday AW, Almarzooq ZI, Anderson CAM, Arora P, Avery CL, Baker-Smith CM, Barone Gibbs B, Beaton AZ, Boehme AK, Commodore-Mensah Y, Currie ME, Elkind MSV, Evenson KR, Generoso G, Heard DG, Hiremath S, Johansen MC, Kalani R, Kazi DS, Ko D, Liu J, Magnani JW, Michos ED, Mussolino ME, Navaneethan SD, Parikh NI, Perman SM, Poudel R, Rezk-Hanna M, Roth GA, Shah NS, St-Onge M-P, Thacker EL, Tsao CW, Urbut SM, Van Spall HGC, Voeks JH, Wang N-Y, Wong ND, Wong SS, Yaffe K,

Palaniappan LP. 2024. 2024 heart disease and stroke statistics: a report of US and global data from the American heart association. *Circulation* **149(8)**:e347–e913 DOI 10.1161/cir.0000000000001209.

Myronenko A. 2018. 3D MRI brain tumor segmentation using autoencoder regularization. In: *International MICCAI Brainlesion Workshop*. Cham: Springer, 311–320.

Oktay O, Schlemper J, Le Folgoc L, Lee M, Heinrich M, Misawa K, Mori K, McDonagh S, Hammerla NY, Kainz B, Glocker B, Rueckert D. 2018. Attention U-Net: learning where to look for the pancreas. ArXiv DOI 10.48550/arXiv.1804.03999.

Ramesh A, Dhariwal P, Nichol A, Chu C, Chen M. 2022. Hierarchical text-conditional image generation with CLIP latents. ArXiv DOI 10.48550/arXiv.2204.06125.

Rombach R, Blattmann A, Lorenz D, Esser P, Ommer B. 2022. High-resolution image synthesis with latent diffusion models. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. Piscataway: IEEE, 10684–10695.

Ronneberger O, Fischer P, Brox T. 2015. U-Net: convolutional networks for biomedical image segmentation. In: *Medical Image Computing and Computer-Assisted Intervention–MICCAI 2015: 18th International Conference, Munich, Germany, October 5–9, 2015, Proceedings, Part III 18*. Cham: Springer, 234–241.

Saharia C, Chan W, Saxena S, Li L, Whang J, Denton EL, Ghasemipour K, Gontijo Lopes R, Karagol Ayan B, Salimans T, Ho J, Fleet DJ, Norouzi M. 2022. Photorealistic text-to-image diffusion models with deep language understanding. *Advances in Neural Information Processing Systems* **35**:36479–36494 DOI 10.52202/068431-2643.

Samak ZA. 2025. Multi-type stroke lesion segmentation: comparison of single-stage and hierarchical approach. *Medical & Biological Engineering & Computing* **63(4)**:975–986 DOI 10.1007/s11517-024-03243-4.

Sarvamangala D, Kulkarni RV. 2022. Convolutional neural networks in medical image understanding: a survey. *Evolutionary Intelligence* **15(1)**:1–22 DOI 10.1007/s12065-020-00540-3.

Schlemper J, Oktay O, Schaap M, Heinrich M, Kainz B, Glocker B, Rueckert D. 2019. Attention gated networks: learning to leverage salient regions in medical images. *Medical Image Analysis* **53(7639)**:197–207 DOI 10.1016/j.media.2019.01.012.

Siddique N, Paheding S, Elkin CP, Devabhaktuni V. 2021. U-Net and its variants for medical image segmentation: a review of theory and applications. *IEEE Access* **9**:82031–82057 DOI 10.1109/access.2021.3086020.

Sohl-Dickstein J, Weiss E, Maheswaranathan N, Ganguli S. 2015. Deep unsupervised learning using nonequilibrium thermodynamics. In: *International Conference on Machine Learning*. PMLR, 2256–2265.

Song J, Meng C, Ermon S. 2020. Denoising diffusion implicit models. ArXiv DOI 10.48550/arXiv.2010.02502.

van den Oord A, Vinyals O, Kavukcuoglu K. 2017. Neural discrete representation learning. In: *31st Conference on Neural Information Processing Systems (NIPS 2017), Long Beach, CA, USA*.

Vu Quoc H, Tran Le Phuong T, Trinh Xuan M, Dinh Viet S. 2023. LSegDiff: a latent diffusion model for medical image segmentation. In: *Proceedings of the 12th International Symposium on Information and Communication Technology*. New York: ACM, 456–462.

Wolleb J, Sandkühler R, Bieder F, Valmaggia P, Cattin PC. 2022. Diffusion models for implicit image segmentation ensembles. In: *International Conference on Medical Imaging with Deep Learning*. Westminster: PMLR, 1336–1348.

Wu J, Fu R, Fang H, Zhang Y, Yang Y, Xiong H, Liu H, Xu Y. 2024a. MedSegDiff: medical image segmentation with diffusion probabilistic model. In: *Medical Imaging with Deep Learning*. Westminster: PMLR, 1623–1639.

Wu J, Ji W, Fu H, Xu M, Jin Y, Xu Y. 2024b. MedSegDiff-V2: diffusion-based medical image segmentation with transformer. *Proceedings of the AAAI Conference on Artificial Intelligence* **38(6)**:6030–6038 DOI 10.1609/aaai.v38i6.28418.

Xing Z, Wan L, Fu H, Yang G, Zhu L. 2023. Diff-UNet: a diffusion embedded network for volumetric segmentation. ArXiv DOI 10.48550/arXiv.2303.10326.

Xue Y, Xu T, Zhang H, Long LR, Huang X. 2018. SegAN: adversarial network with multi-scale L1 loss for medical image segmentation. *Neuroinformatics* **16(3–4)**:383–392 DOI 10.1007/s12021-018-9377-x.

Yalçın S, Vural H. 2022. Brain stroke classification and segmentation using encoder-decoder based deep convolutional neural networks. *Computers in Biology and Medicine* **149(3)**:105941 DOI 10.1016/j.compbiomed.2022.105941.

Yu F, Seff A, Zhang Y, Song S, Funkhouser T, Xiao J. 2015. LSUN: construction of a large-scale image dataset using deep learning with humans in the loop. ArXiv DOI 10.48550/arXiv.1506.03365.

Zaman FA, Jacob M, Chang A, Liu K, Sonka M, Wu X. 2024. Denoising diffusions in latent space for medical image segmentation. ArXiv DOI 10.48550/arXiv.2407.12952.

Zhang D, Confidence R, Anazodo U. 2022. Stroke lesion segmentation from low-quality and few-shot MRIs via similarity-weighted self-ensembling framework. In: *International Conference on Medical Image Computing and Computer-Assisted Intervention*. Cham: Springer, 87–96.

Zhang Z, Liu Q, Wang Y. 2018. Road extraction by deep residual U-Net. *IEEE Geoscience and Remote Sensing Letters* **15(5)**:749–753 DOI 10.1109/lgrs.2018.2802944.

Zhou SK, Greenspan H, Shen D. 2023. *Deep learning for medical image analysis*. Cambridge: Academic Press.

Zhou Z, Rahman Siddiquee MM, Tajbakhsh N, Liang J. 2018. UNet++: a nested U-Net architecture for medical image segmentation. In: *Deep Learning in Medical Image Analysis and Multimodal Learning for Clinical Decision Support: 4th International Workshop, DLMIA 2018, and 8th International Workshop, ML-CDS 2018, Held in Conjunction with MICCAI 2018, Granada, Spain, September 20, 2018, Proceedings 4*. Cham: Springer, 3–11.